Geometric Reconstruction and Mechanical Analysis of the B&M Clamshell Lap Restraint

u/lucky_3838[1]

Abstract

The locking mechanism of a B&M Hyper-family clamshell lap restraint is reconstructed from public drawing captures, photographs and original CAD contours. Nine reconstructed pawl–tooth contact configurations satisfy frictionless in-plane equilibrium. Nominal stresses at sampled sections and a reduced torsional model connect pad forces to both locking sides across seven closure pairs. With assumed pawl/tube yield strengths of 355 MPa, one reference configuration gives nominal yield thresholds of 4.66 kN upwards and 5.03 kN forwards at the pad centre. Numerical readings recovered from 2898 frames of a chest-mounted Mako video inform occupant-load scenarios; interpolated values are excluded. The most demanding selected scenario assumes seated braking and a pad force directed 60° above the forward horizontal, applied at the outer end of the modelled contact segment. Calculated resultant forces are 2.50 kN for an assumed occupant mass of 150 kg and 3.34 kN for 200 kg. For that direction and configuration, the force at which a calculated nominal section stress first reaches the assumed 355 MPa yield strength is 3.25 kN. The 150 kg force is below this threshold and the 200 kg force above it. This comparison does not predict fracture or release. Actual assembly capacity, passenger mass limits and service adequacy remain unresolved.

Keywords: lap restraint, clamshell, B&M, pawl, unilateral contact, CAD reconstruction, load transfer

Independent working paper. Research revision: 19 October 2026.
Earlier published edition: 17 October 2026.

Independence and scope. I have no professional, contractual or institutional affiliation with B&M. This project grew out of an earlier technical interest; it is not part of the investigation into the Six Flags Over Georgia accident and does not seek to establish its cause. Calculations describe idealised models based on reconstructed geometry: their decimal precision does not establish measurement accuracy or validate the physical restraint. They do not constitute a certified capacity, maintenance instructions or operating criteria. Drawing captures come from material shown in Ryan The Ride Mechanic's video, where the marking *confidentiel* appears; B&M did not provide these captures directly for this study.

[1] I am a university lecturer and researcher in applied physics, with an engineering background, a PhD and experience in numerical modelling. This independent study grew out of my personal interest in roller-coaster engineering.

Contents

1 Introduction and research questions 3

2 Historical context and the containment concept 4

3 Publicly available knowledge and the scope of this contribution 7

4 Evidence and reconstruction methods 8

5 Geometric reconstruction and component identification 10

6 Closing kinematics, engagement positions and release 14

7 Axial architecture, supports and continuity of the load path 18

8 Contact, seating and opening restraint 21

9 Internal forces and conditional strength assessments 24

10 From occupant dynamics to restraint loading 27

11 From thigh contact to internal loads 29

12 Conditional nominal load thresholds 32

13 Assembly-level scenarios with a common arm rotation 37

14 Numerical verification and model sensitivity 43

15 Published ride measurements and conditional demand histories 45

16 Discussion 55

17 Conclusions and further work 58

Author note and contributions 59

Appendix A Seated contact configurations 63

Appendix B Selected source video frames 65

1. Introduction and research questions

From the rider's seat, a roller coaster lap bar appears straightforward: a pivoting arm, a padded restraint and a series of clicks as it closes. That experience conceals several distinct mechanical problems. The assembly must accommodate different body dimensions, maintain containment during the ride, transmit forces to the vehicle structure and allow release during loading and unloading. Understanding the lock requires considering these functions together while keeping their roles distinct. A rigid bar may still provide inadequate geometric containment for a particular posture; a sensor may confirm position without measuring component strength; and several pawls may share structural components.

This study examines the clamshell lap restraint used in Bolliger & Mabillard's (B&M) Hyper family through documentary research and geometric reconstruction. The case examined here is based on photographs attributed to Mako, captures of an assembly drawing and original CAD reconstructions. Its scope is the architecture represented by that material. External resemblance between rides in the same family is not taken as evidence of identical parts, revisions or service conditions.

Figure 1 connects the rider-facing arrangement to the mechanism studied here. The pad and long arm are visible above the seat; the locking assembly sits at the arm's pivot, where pad forces become moments resisted by the pawls and toothed sectors. The photographs provide spatial context for the reconstructed sections that follow.



(a) Seats, pads and raised restraint arms.



(b) Locking assembly at the arm pivot.

Figure 1: Photographic context attributed to Mako. Photographs reposted by Velocity Rides in the Facebook group Roller Coaster Enthusiasts Club; the original photographer has not been identified.[21] Panel (b) is processed and upscaled; it illustrates visible components without adding dimensional evidence.

The central question is how a force applied to the pad reaches the locking assembly and how its contacts prevent opening rotation. Five objectives follow: identify the bodies and their joints; reconstruct the discrete closing sequence; find contact configurations consistent with equilibrium; convert applied moments into internal forces and nominal stresses; and estimate conditional load thresholds while identifying the observations still needed to assess the physical assembly's capacity.

The study connects these stages through a traceable chain of data and assumptions. Original analytical drawings allow a more detailed examination of components and contact states than a verbal description. Results per unit applied moment avoid fixing an unknown rider load at the outset. Scenarios for force magnitude, contact position and direction show which results change with the external geometry. Throughout the analysis, legible dimensions are distinguished from scaled measurements, functional interpretations and parameters selected for sensitivity analysis.

The strength assessment calculates contact reactions, fastener-group loads, nominal stresses at accepted sections, and bending or shear scenarios for pins and tubes. It also follows upward, forward and combined forces to the first included nominal stress criterion, with sensitivity to material assumptions, closure position and load sharing. These conditional thresholds are developed in Sections 12 and 13. They do not establish the restraint's physical maximum load: that would require verified materials, treatments, radii, clearances, supports, connections, wear and dynamic loads, together with assessment of failure modes that the model does not represent.

Lock strength is also distinguished from rider containment. The latter requires the actual seat and pad surfaces, anthropometry, posture and body kinematics. The equivalent arm used here reduces part of that interaction to a load-transfer model. It neither reproduces distributed pressure on the rider's legs nor constitutes a biomechanical model.

The present research revision also asks how published ride-motion records can inform those prescribed loads. Section 15 reviews experimental evidence, recovers same-frame numerical Mako readings, identifies their source video timestamps, and compares explicitly different occupant-contact hypotheses. Motion measurements and calculated resistance remain separate evidence streams.

2. Historical context and the containment concept

Bolliger & Mabillard was founded in 1988 by Walter Bolliger and Claude Mabillard and is based in Monthey, Switzerland. Its current corporate presentation distinguishes design, engineering, fabrication coordination, testing and support throughout an installation's service life. This description places the restraint within an engineered and integrated system, but does not identify the manufacturer of each component in the assembly studied here.[1]

2.1. The emergence of the Hyper generation

The parks' own publications document Apollo's Chariot and Raging Bull in 1999. Busch Gardens celebrated Apollo's Chariot's twentieth anniversary in 2019; Six Flags identifies Raging Bull as a B&M ride introduced in 1999.[3, 4] These are installation dates, not dates of a particular revision of the internal lock.

B&M's current catalogue presents Hyper as a family focused on speed, large drops and airtime, emphasizing the sense of freedom associated with its lap bar. The same page lists the installations in Table 1. Continuity within a product family provides context for the case study; it does not establish that restraint components are interchangeable between trains.[2]

Figure 2 shows that ride context at Mako. The large-scale track profile motivates the discussion of rider loading, while the close views in Figure 1 locate the restraint and its lock within the vehicle. Neither view supplies a force history or identifies the revision of the internal mechanism.



Figure 2: Mako at SeaWorld Orlando, 10 March 2021: lift hill, first drop and lakeside hills. Photograph by ThrillZing, via Wikimedia Commons, licensed under CC BY 4.0.[22] Reproduced without image changes.

Table 1: Selected milestones placing the case study in context. Years refer to the installations identified by the sources, not restraint revision numbers.

Year	Installation	Primary source
1999	Apollo's Chariot; Raging Bull	Respective parks
2001	Nitro	B&M catalogue
2006	Goliath, Six Flags Over Georgia	B&M catalogue
2009	Diamondback	B&M catalogue
2012	Shambhala	B&M catalogue
2016	Mako	B&M catalogue
2020	Candymonium	B&M catalogue

2.2. What the original patent establishes

US6287211B1 names Walter Bolliger and Claude Mabillard as inventors, with priority on 11 January 1999, US filing on 22 October 1999 and publication on 11 September 2001. Figure 3 illustrates its occupied seat and shaped retention element. The drawing connects the pivoting bar and pad to the seat's containment geometry.[5]

For adjustment, the patent mentions a ratchet box acting on a toothed wheel but omits its internal details as a known mechanism. It therefore supports the containment concept, without supplying the pawl geometry or material specifications required here. It is not treated as a production drawing of Mako's restraint.[5]

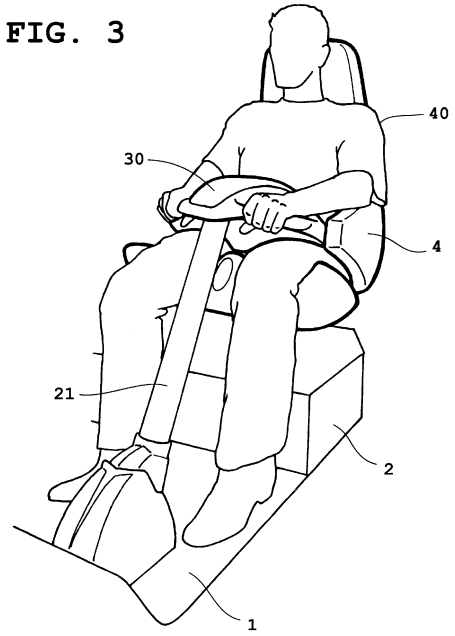


Figure 3: Occupied seat–restraint arrangement from US6287211B1, original FIG. 3.[5] Numerals 21 and 30 identify the bar and retention element in that patent; they are unrelated to this article's component identifiers. The original drawing is retained, with the publication header and surrounding whitespace omitted.

2.3. Seat geometry as part of the problem

Three levels are useful in modelling. First, the seat, lateral supports and pad define the space available to the rider and constrain movement. Second, the joint and lock determine the adjustable position of the bar. Third, the load-bearing components preserve that position under load. These levels interact, but checking the third does not replace assessment of the first two.

Mako's accessibility guide requires an upright posture with the back and shoulders against the seat back, alongside other functional requirements.[12] This supplies operating-posture context and reinforces the relevance of the interaction between rider and vehicle, but gives neither an anthropometric design model nor dimensions of the locking assembly. No design loads, pad profiles or contact-force allocation are inferred from those requirements here.

2.4. Industrial supply: limits of attribution

Valente SAS lists Bolliger & Mabillard among its commercial references and describes machining, assembly and inspection capabilities.[9, 10] This documented relationship does not identify the supplier of each pawl or establish who produced a particular batch of trains. Statements about track materials or a workshop's welding practices likewise cannot be transferred to lock components. Assigning an alloy, treatment or inspection requirement to those components would require part-specific documentation.

3. Publicly available knowledge and the scope of this contribution

The public material consulted addresses different parts of the problem: patents on containment and locking mechanisms; general manufacturer and park documentation; simplified academic analyses; accounts of operation and maintenance; and photographs of components. The existence of these categories does not imply that a complete public engineering record exists for the clamshell studied here. The review is organized around the questions each source can answer.

3.1. Patents: mechanism, monitoring and function

Disney patent US7947943B2, published in 2011 from a 2007 application, addresses indirect sensing of a pawl-and-ratchet assembly's operation. Its conceptual relevance is the distinction between the overall position of a restraint and a pawl's movement into engagement, with displacement amplification proposed to make that movement observable by a sensor.[6] This distinction helps formulate questions about monitoring. It does not establish which sensor a B&M Hyper uses or what the device visible in the photographs measures.

Another Disney patent, US8132514B2, published in 2012, describes a lap bar that locks both in the ride position and in a raised position where it can serve as a support during boarding. It uses a toothed wheel, pawls, cams and followers to provide distinct functions.[7] The example illustrates why a component near a travel limit should not automatically be called a *locking pin*. Its role must be established from the contacts and kinematics of the particular assembly. That patent is not used as a substitute drawing of the B&M mechanism.

The illustrated embodiment also mounts both pawls on a common shaft so that they rotate together, while describing redundancy from the two contacts.[7] It supplies a concrete example of the distinction between multiple locking contacts, independent motion and shared components. This is a feature of that patent embodiment, rather than an identification of the B&M arrangement.

These sources contribute terminology and design questions, rather than the missing dimensions of the case study. They are not evidence that a feature was implemented on a particular ride. Comparing patents helps prevent a documented feature of one design from becoming an assumed property of another through visual resemblance alone.

3.2. An earlier calculation and its limitations

Greathead, Taylor and Rai published a physics exercise in the undergraduate *Journal of Physics Special Topics* in 2024. Their model is inspired by a B&M clamshell, assumes two parallel mechanisms with combined effective pawl dimensions of $4 \times 1 \times 8$ cm, and applies Euler buckling

with $E = 200$ GPa and effective-length factor $K = 2$. It obtains an internal load of about 250 kN and converts this to approximately 35 kN at the bar.[8]

Those values belong to the assumed model. The article reports no dimensional survey of an actual B&M mechanism, identification of its alloy or assembly load test. Its 35 kN result is therefore not adopted as a reference capacity here. The governing failure mode also requires justification: buckling, yielding, local bending, contact and loss of engagement are distinct checks. The article's biomechanical comparison lies outside this study.[8]

The present objective is narrower: use reconstructed contours to determine admissible contact forces and nominal stresses at documented sections, while separating recovered geometry from material and load scenarios. Better geometry does not remove the need to validate those scenarios.

3.3. Measured rider motion and the missing contact forces

Figueroa Jacinto, Hybart and Zendler's exploratory field study used an IMU-based motion-capture suit on one participant across five amusement devices. It recorded body-segment and centre-of-mass kinematics, while explicitly lacking direct ride-acceleration and rider–ride interaction-force measurements.[30] This provides a complementary example of the distinction between measuring motion and identifying the restraint load. It supplies no force history or strength validation for the B&M mechanism considered here.

3.4. Public commentary and visual evidence

Public commentary can suggest questions about operation and maintenance, but does not establish component geometry, materials or capacity. A description of two ratchets, for example, would not establish structurally independent load paths. The video used for the present reconstruction provides relevant images alongside speculative commentary.[20] The images are used for geometric interpretation; commentary about an incident is not treated as a mechanical finding.

3.5. The specific gap addressed

Among the primary sources verified for this edition, no public account was found that combines a dimensioned breakdown of this assembly, its contact sequence, load sharing between sides, material properties and capacity testing. This describes the material consulted; it is neither a claim of universal absence nor a priority claim for this work.

The contribution presented here combines vector drawings of the sections and closing sequences, a component identification table, unilateral equilibrium at seated contact states, and a parametric load analysis connecting external forces to internal responses. The equations, assumptions and representative results are stated so that the model can be reassessed when better dimensions become available.

4. Evidence and reconstruction methods

4.1. Sources and levels of confidence

The analysis combines patents and other cited public documents, photographs and video captures, and original CAD reconstructions. Direct observations, legible drawing information, geometric interpretations and selected modelling assumptions are distinguished throughout. Reconstructed

contours and calculations are analytical results, rather than manufacturer-authenticated drawings or test data. A photograph may document a visible feature while the identity or revision of the assembly remains uncertain.

4.2. Patent, assembly drawing and video capture

The original patent provides conceptual and historical context. The sheet headed *Locking Device Assembly Part 22*, visible in video captures, is a separate source. Its A–A and B–B sections and complementary views support a more detailed architectural interpretation, but a capture is not an original drawing under revision control. The apparent industrial authorship of the sheet, the provenance of its photograph and the attribution of the assembly to a ride are kept distinct.

The captures show the marking *confidentiel*. They were taken from material shown in Ryan The Ride Mechanic's video; B&M did not provide these captures directly for this study.[20] Public availability identifies how the material was encountered, without authenticating the sheet's revision, completeness or authorization for distribution. The CAD figures in this work are analytical reconstructions.

The material reproduced in the video is credited on screen to `@Crazybird199`. At approximately 5:44, Ryan also states that the following pictures were supplied by that subscriber.[20] This credits the contribution and its route into the video; it does not establish authorship of the underlying technical drawing. The commentary identifies the photographs as showing a Hyper other than Goliath, but does not explicitly establish Mako in the relevant passage. The photographs are therefore described here as *material attributed to Mako*; that attribution is not independently authenticated. A legible part number helps trace a component, but does not by itself establish the ride, manufacturing date or latest revision.

Five selected video frames are reproduced in Appendix B, retaining the presenter inset and visible credits. They show the drawing sheet, loose components and exposed mechanism views, so that the visual sources can be compared with the analytical reconstruction.

The CAD figures are original analytical reconstructions, rather than enhanced scans presented as original documents. The contextual photographs in Figure 1 have a separate provenance: a Facebook post by Velocity Rides in the group Roller Coaster Enthusiasts Club, identified through a capture of the post.[21] The original photographer has not been established, and the post's surrounding incident commentary is not used as a technical finding. The processed close-up is identified as such. Image interpolation may improve visual readability, but cannot recover dimensions or details unresolved in the source; it provides no additional metrological evidence.

4.3. Reconstructing the two sides

The assembly drawing was used to distinguish contours in the principal section plane from components behind it. Separate reconstructions of A–A and B–B identify the toothed sectors, pawls and release members; the closing sequences represent nominal successive poses. The mechanism is analysed from these drawing-informed contours rather than from repeated teeth extrapolated from a single exposed fragment in a photograph.

A3 and B3 probably share part reference 2423: it is legible in B–B and inferred in A–A by correspondence. Because their silhouettes were traced separately, small discrepancies are not presented as evidence of different physical designs. The proposed identity supports a common topology, but does not determine unmeasured axial spacing, fits or support surfaces.

4.4. Scale and dimensional confidence

Dimensions recovered by scaling the drawing are estimates from the reconstruction rather than written specifications. Absolute scale depends on assumed fastener dimensions and proportional checks, rather than an authenticated calibration dimension. The reconstruction uses an approximately 60 mm outside tube diameter, an approximately 5 mm nominal wall, and estimated axial thicknesses of 16 mm for pawls and 32 mm for toothed sectors. These values define the model; their apparent consistency does not make them manufacturer dimensions.

The 780 mm pivot-to-junction distance, 250 mm equivalent pad segment and 98-degree angle were estimated from a perspective-corrected photograph. Scale was set using the shaft bore interpreted as 32 mm and checked against a tube diameter close to 60 mm. Agreement is a useful internal consistency check. Both references inherit uncertainty from the same reconstruction and do not remove possible depth differences between mechanism and pad. Sensitivity ranges selected later are scenarios, not statistical confidence intervals.

4.5. Goliath context and separation from the case study

The idea of studying this mechanism predates the Goliath incident of 12 October 2026. The study resumed when newly available captures made a more detailed reconstruction possible. In its 15 October statement, the state authority OCI reported that its investigation remained open and the ride remained out of service. It also stated that the 16 March inspection had recorded no deficiencies.[11] This documentary status neither establishes a mechanical cause nor makes photographs of another Hyper evidence of the incident.

The analysis presented here uses no incident images and does not reconstruct the occupants' posture. Fracture, loss of engagement, release actuation and inadequate containment are different mechanical categories requiring different evidence. Solving equilibrium in the CAD reconstruction does not distinguish between them in a real event.

5. Geometric reconstruction and component identification

5.1. Drawing conventions

Figure 4 preserves the distinction between geometry visible in the principal section plane and components behind it. The toothed sectors attached to the restraint arm are shown in blue; the pawls in orange; the release linkage in green; and pin A9 in purple. A dashed line denotes a component outside the principal plane or an explicitly identified approximate representation. It does not, by itself, indicate a thinner or less structurally significant component. Hatching identifies solid sections where the drawing distinguishes them, while centre lines locate the joints.

Local identifiers A1–A13 and B1–B6 maintain correspondence between figures and tables. They are not catalogue numbers. The main shaft is identified by its hatched annular section; it is distinguished from the immediately adjacent circular housing outline. A12 and B6 identify the same physical component. A3 and B3, by contrast, are two separate pawls on opposite sides that probably share a part number.

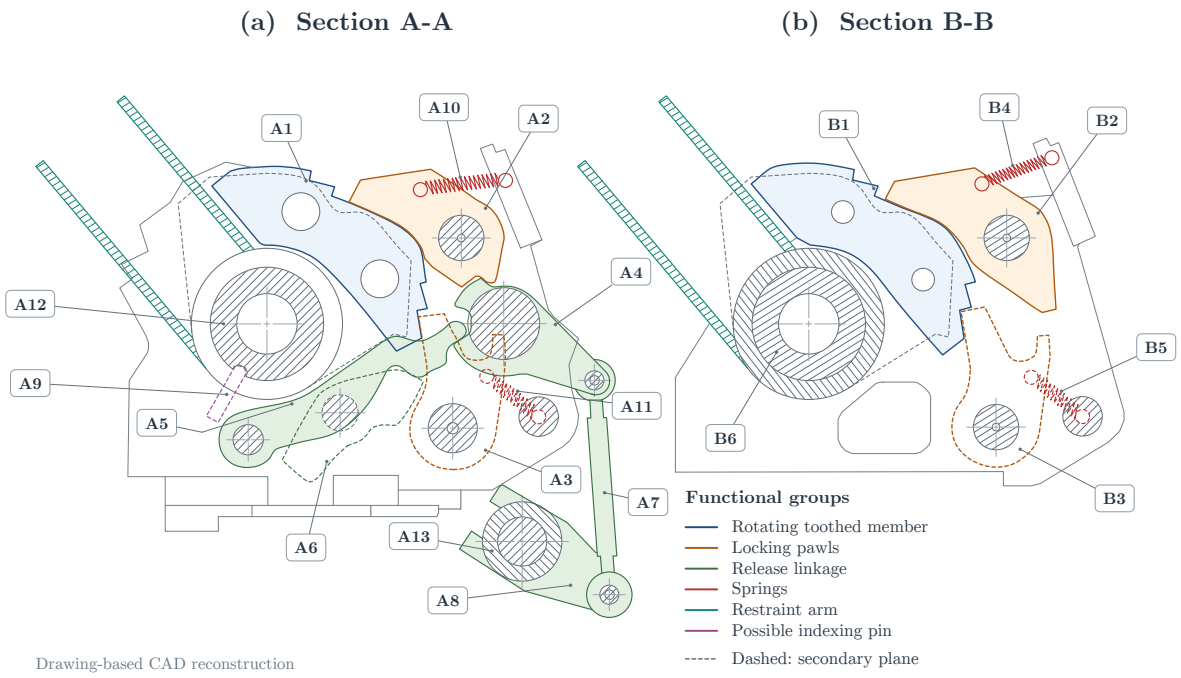


Figure 4: Analytical vector reconstruction of sections A–A and B–B from the assembly-drawing captures. The views correspond to opposite sides of the restraint arm. Local identifiers refer to Table 2; colours are analytical annotations. The dashed grey carrier outline is approximate.

Table 2: Functional identification and B&M part references transcribed from the assembly drawing. An asterisk denotes an assignment inferred by correspondence. N/A: reference not identified.

ID	Section	Functional identification	B&M ref.
A1	A-A	Rotating toothed sector	3974
A2	A-A	Foreground pawl	2421
A3	A-A	Rear-plane pawl	2423*
A4	A-A	Upper release rocker	3969
A5	A-A	Foreground release lever	3971
A6	A-A	Rear-plane release lever	2416
A7	A-A	Connecting link	N/A
A8	A-A	Lower release lever	N/A
A9	A-A	Possible full-closure indexing pin	N/A
A10	A-A	Upper pawl spring	2429
A11	A-A	Lower pawl spring, rear plane	5167
A12	A-A	Main shaft, hatched section	5832
A13	A-A	Release-system pivot shaft	N/A
B1	B-B	Rotating toothed sector	3973
B2	B-B	Foreground pawl	9455
B3	B-B	Rear-plane pawl	2423
B4	B-B	Upper pawl spring	5157
B5	B-B	Lower pawl spring, rear plane	5167
B6	B-B	Main shaft; shared with A12	5832

*A3: probable reference 2423, inferred from correspondence with B3; the reference is readable in B–B but is not shown for A3 in A–A.

Identifiers A12 and B6 represent one physical shaft, reference 5832. A3/B3 are two separate pawls with a probable shared reference, 2423; small differences between their independently traced outlines do not establish a physical design difference. A11/B5 share reference 5167 without being consolidated into a single unit.

References for A7, A8, A9 and A13 have not been identified. A13 is the release-system pivot shaft.

A9 is provisionally interpreted as a full-closure indexing pin. No retention contribution is assigned to it in the current model. A13 is retained to explain release kinematics. These functions do not imply that the components carry no reactions.

The 24 and 26 mm pin diameters used in sensitivity calculations remain hypothetical; circles in the reconstruction do not establish the physical pin construction.

Functional names are descriptive. A part number alone does not establish material, treatment, stiffness or capacity. Drawing number, sheet, revision and callout location for each reading remain necessary for complete traceability.

5.2. Bodies, joints and load path

Sector A1 or B1 rotates with the shaft and restraint arm. Each pawl has its own pivot and can follow the profile during closure. When an opening load acts on a seated configuration, contact transfers force from the sector to the pawl; the pawl joint transfers that force to the structure. For the complete assembly, the load path also includes the arm-to-shaft connection, carrier, sector fasteners and support interfaces.

The approximate dashed grey outline represents the shaft extension to which the sectors are bolted. It does not represent a rear toothed profile or a stationary housing stop acting on the pawl. The two visible fasteners are described as having heads approximately 20 mm in diameter and an internal threaded diameter of approximately 12 mm. These measurements do not identify a nominal thread size, strength class or effective area. Introducing an independent support reaction at the centre of an isolated sector would incorrectly bypass this connection.

The springs promote pawl tracking and return. The green linkage withdraws the pawls. These functions are identified from the represented motion, but free lengths, stiffnesses, preloads and actual anchor positions remain unavailable for force calculations. A spring coil drawn in each pose is not an experimental force–displacement curve.

5.3. Recovered dimensions and selected parameters

Table 3: Principal dimensions used in the analysis. The classification describes the origin of each value, not a manufacturing tolerance.

Component	Working value	Provenance and scope
Arm tube	$D \simeq 60$, $t \simeq 5$ mm	Measured from drawing B–B; not a written specification.
Hatched shaft section	$D \simeq 60$, $d \simeq 32$ mm	Ideal annulus corresponding to circles in the CAD reconstruction.
Pawls	$t \simeq 16$ mm	Approximate scaling of the low-resolution axial view.
Toothed sectors	$t \simeq 32$ mm	Same procedure; compatible with two pawl bands, without resolving the axial stack.
External arm geometry	$L \simeq 780$ mm	Pivot-to-pad-junction distance estimated from a rectified photograph.
Equivalent contact segment	$a \simeq 250$ mm; $\beta \simeq 98°$	Ideal side-view thigh-contact segment and its interior angle to the direction towards the pivot.
Pawl pivot pins	$D = 24$ or 26 mm	Scenarios based on DXF circles; the physical object and actual diameter are unverified.
Pin support spans	32, 48, 64 mm	Selected sensitivity values, not recovered dimensions.

The side profile and axial thickness are different inputs. An accurately traced outline can provide in-plane lever arms, but does not establish the load-bearing area out of that plane. Local thickness reductions due to machining, chamfers or bushings are also unknown. Adopting $t = 16$ mm defines an ideal prismatic section; the photograph of thick components does not verify constant thickness.

5.4. Coordinate frames and sign conventions

Contact calculations use the coordinate frame of each DXF pose, with origin O at the shaft axis, x to the right and y upwards. The planar cross product is defined as $\mathbf{a} \times \mathbf{b} = a_x b_y - a_y b_x$,

positive anticlockwise. The magnitude $M \geq 0$ denotes the *clockwise* opening moment applied to the sector; contact must therefore supply an anticlockwise moment $+M$ for equilibrium. Geometric coordinates are in mm and applied moments are reported in N m.

The initial load envelope in Section 11 uses a tube-aligned local frame. The directional-load scenarios in Sections 12 and 13 use an assumed seat frame with x forward and z upward. Force components expressed in these different frames cannot be added directly. A global free-body diagram requires rotation into a common frame and transfer to a common reference point, retaining the associated couples. Scalar moments allow scenario coupling without implying that the complete spatial assembly has already been reconstructed.

6. Closing kinematics, engagement positions and release

6.1. A shared angular reference and two different profiles

The A–A and B–B sequences in Figures 5 and 6 use the open assembly as a nominal 0° reference. They show how the rotating toothed profiles encounter the foreground and rear-plane pawls at successive drawn positions. The common reference permits comparison between the two sides, but is neither an absolute orientation relative to the passenger seat nor a manufacturer adjustment dimension. Each pawl has its own pivot and can follow the profile during closing. The subsequent contact analysis checks individual configurations rather than simulating the complete motion with elasticity, preload, friction and impacts.

Table 4: Nominal click positions in the reconstructions, ordered by angle. The pawl column identifies the local seated configuration subsequently analysed; it does not establish simultaneous load sharing with the opposite side.

Angle	Section	Label within that section	Pawl analysed
24°	A–A	First click	A2, foreground
27°	B–B	First click	B2, foreground
29°	A–A	Second click	A3, rear plane
32°	B–B	Second click	B3, rear plane
33°	A–A	Third click	A2, foreground
36°	B–B	Third click	B2, foreground
38°	A–A	Fourth click	A3, rear plane
40°	B–B	Fourth click; final closure	B3, rear plane

With the shared nominal reference, their order is

$$A24 \longrightarrow B27 \longrightarrow A29 \longrightarrow B32 \longrightarrow A33 \longrightarrow B36 \longrightarrow A38 \longrightarrow B40.$$

The labelled intervals are 3, 2, 3, 1, 3, 2 and 2°. This describes eight interleaved positions in the drawings. It is not a measurement of eight distinguishable sounds, proof of eight globally stable assembly states, or a dispatch criterion. Motion speed, rebound, approximate angle labels and several moving components could alter the acoustic signal; no such signal was measured here.

The A–A sequence also includes a nominal final panel at 40°, beyond its fourth click at 38°. The final closure reference is nominal; it is not a separately verified manufacturing angle. This

extra panel does not introduce a fifth A–A click. The reconstructed A–A profiles at 38° and 40° reach the same load-bearing tooth configuration. A9 is provisionally interpreted as a full-closure indexing pin, with no retention contribution assigned in the model. No indexing pin is depicted in the B–B reconstruction.

6.2. From a drawn pose to a loaded configuration

Pawl and spring positions in the sequence panels were adjusted to depict plausible motion. Contact analysis subsequently allows the sector and selected pawl to move towards a compatible seated configuration. Its opening adjustment δ and pawl rotation ϕ are model results rather than measured backlash. The first-touch and equilibrium calculations are developed in Section 8; Section 13 couples reconstructed seats through an assumed torsional transmission network and examines seven closure-position pairs. The actual spatial contact and deformation problem remains unresolved.

6.3. Functional interpretation of release

The green components in A–A depict the release transmission. In the proposed functional interpretation, the rising actuator shoe pushes a roller and moves a system of levers and links that withdraws the pawls from the teeth. Continuous and dashed green traces distinguish foreground and rear-plane components. The visible chain is interpreted through the lower lever A8, its rotation axis A13, connecting link A7 and the upper release members, including A4–A6. These identifications describe the reconstructed linkage rather than establishing every three-dimensional connection or phase relationship.

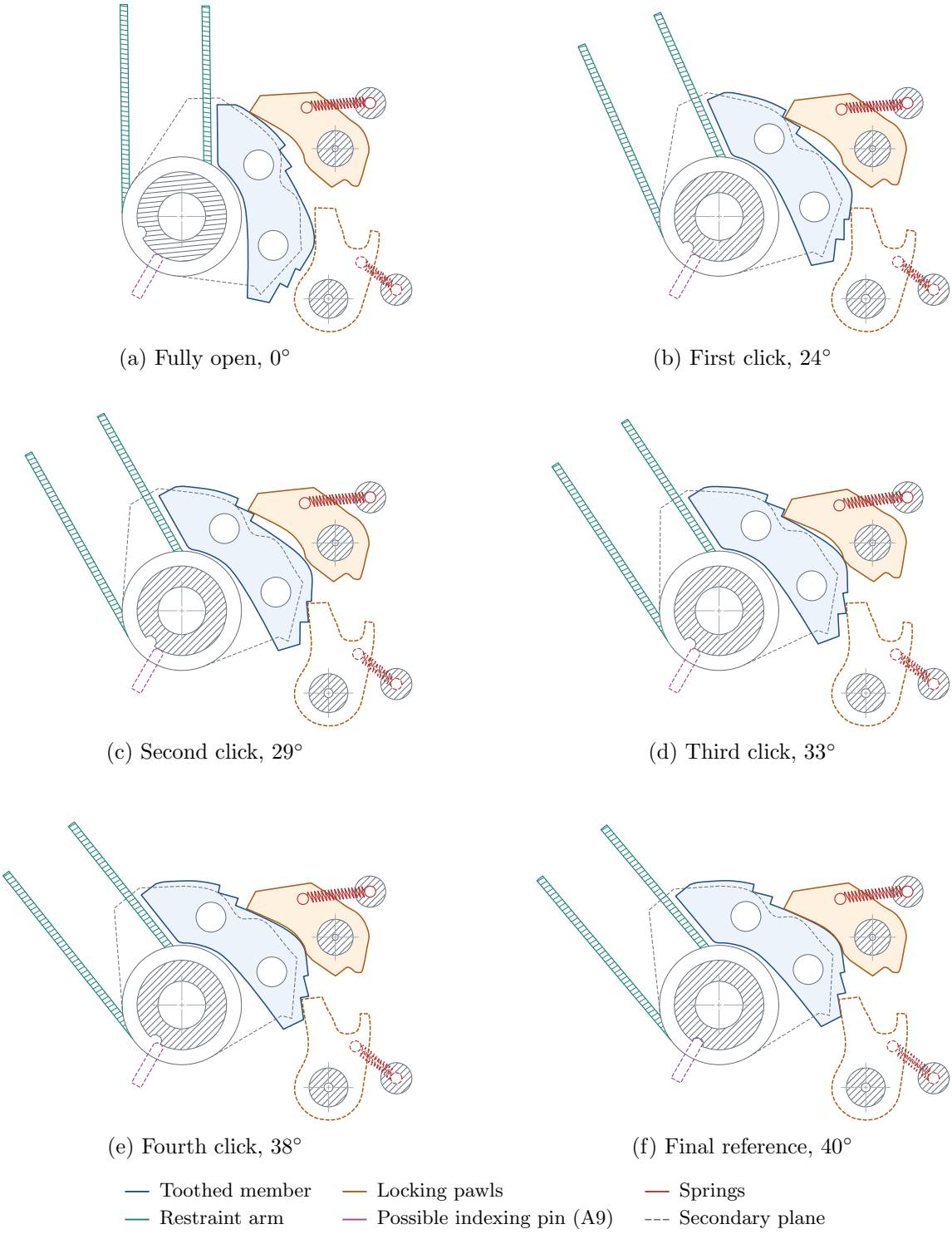
A5 and A6 are associated with the respective foreground and rear-plane pawls. Their separate actions explain why two pawls may be withdrawn individually within a coordinated release system. The drawing does not by itself demonstrate independent actuators, independent energy supplies, or the behaviour following a failure of a shared link. A13 is retained as a release-system rotation axis; its part number is a secondary documentation gap, not an unresolved locking function.

A mechanical release model would require pivot locations in a common frame, contact profiles at the roller and shoe, link lengths, clearances and spring forces. A force diagram for the release position would also differ from the opening-load diagram. No release force, actuator margin or operating procedure is derived from the present geometry.

6.4. Spring function and remaining measurements

The springs bias pawls towards following and engagement. Their force could influence contact continuity, the transient response of a click and the actuation required for release. It should not be inferred from a visually plausible coil drawn at each angle. Stiffness, free length, initial extension, attachment points and effective moment arms remain unknown.

In a later quasistatic model, a measured spring law would give $F_s = k_s(\ell - \ell_0)$ over its appropriate working range, followed by a moment about the pawl pivot. Such a term could be added to the pawl equilibrium with gravity and actuation loads. The current contact solutions instead describe incremental opening load in selected seated configurations, without crediting an unmeasured spring torque as a substitute for geometric retention.



(a) Fully open, 0°

(b) First click, 24°

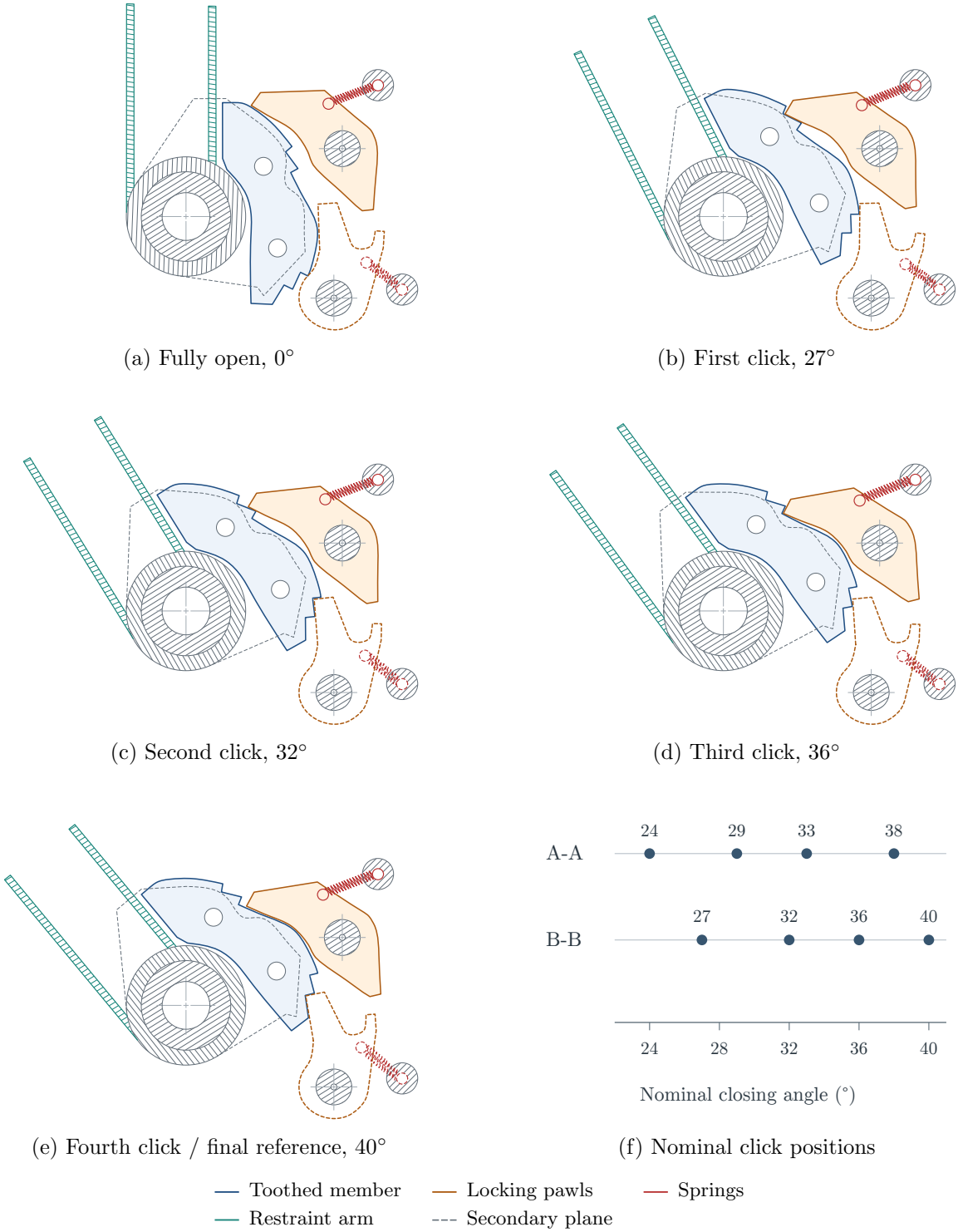
(c) Second click, 29°

(d) Third click, 33°

(e) Fourth click, 38°

(f) Final reference, 40°

Figure 5: Reconstructed A–A closing sequence. Angles are nominal offsets from the open reference. Panels (e) and (f) retain the fourth-click tooth configuration; 40° is the final drawing reference, not a universal occupied position. A9 is provisionally interpreted as an indexing pin, with no retention capacity assigned to it.



(a) Fully open, 0°

(b) First click, 27°

(c) Second click, 32°

(d) Third click, 36°

(e) Fourth click / final reference, 40°

(f) Nominal click positions

Figure 6: Reconstructed B–B closing sequence and comparison of nominal click positions. Agreement of the A–A and B–B final references at 40° is a drawing consistency check, not a measurement of clearances or evidence that all four pawls carry load simultaneously.

7. Axial architecture, supports and continuity of the load path

The drawing's section locator places A–A and B–B on opposite sides of the restraint arm. A12 and B6 identify one continuous main shaft, reference 5832. The views therefore describe different regions of a coupled assembly rather than mechanisms with independently moving arms. Sectors A1 (3974) and B1 (3973) transfer load to that shaft through their fixings and the integral carrier. The dashed grey CAD contour illustrates the carrier approximately; its exact manufacturing geometry has not been reconstructed.

7.1. Pawl identity and the information contained in each view

Reference 2423 is readable for B3 in B–B. Assigning the same reference to A3 is an inference from corresponding shape and function: the available A–A material does not show a reference for A3. The two contours were traced independently, and their small differences are attributed to manual reconstruction. Those discrepancies are therefore not treated as evidence of different physical designs, and exact rigid-copy agreement is not required to identify the probable common part. Numerical geometry remains traceable to each reconstructed view. A shared reference identifies a part design, not one physical pawl extending across both sides.

The four pawl identifiers are A2 (2421), A3 (2423, inferred), B2 (9455) and B3 (2423, transcribed). Approximate working thicknesses are 16 mm per pawl and 32 mm per sector, estimated from the axial drawing. Those values alone do not resolve the stack. In particular, $32 = 2 \times 16$ is compatible with two axial bands but does not establish centring, clearance, band order, effective contact width or support positions. Figure 7 collects the proposed motion and load-transfer connections; its graph deliberately leaves the unresolved axial ordering and support geometry unspecified.

7.2. What can be stated about the supports

Sections and photographs show articulation axes and regions surrounding the pivots. The functional model represents support interfaces between shaft, pivots and frame. Bushing length and material, fit, the number of reaction planes and their separation are not verified. A hatched circular section does not unambiguously distinguish a pin body, bushing, bolt or extension of another member.

The 24 and 26 mm diameters used in the pin sensitivity calculations of Section 9 come from CAD circles. They remain *assumed solid-pin diameters for sensitivity*, rather than OEM dimensions of the A3/B3 pivot pins. Probable pawl identity does not prove that those separate circles depict the same physical object or justify selecting a real diameter between them. Single or double shear also cannot be selected without resolving the support interfaces.

7.3. Spatial equilibrium without prescribing load sharing

For a selected quasistatic free body, external forces act at their transfer points. With all position vectors referred to one origin and applied couples $\mathbf{m}_j$,

$$\sum_j \mathbf{F}_j = \mathbf{0}, \qquad \sum_j \mathbf{r}_j \times \mathbf{F}_j + \sum_j \mathbf{m}_j = \mathbf{0}.$$

For an isolated sector, pawl contact forces are balanced by the carrier connection; no fictitious independent support is added at the shaft centre. For the shaft and carrier, their actual supports

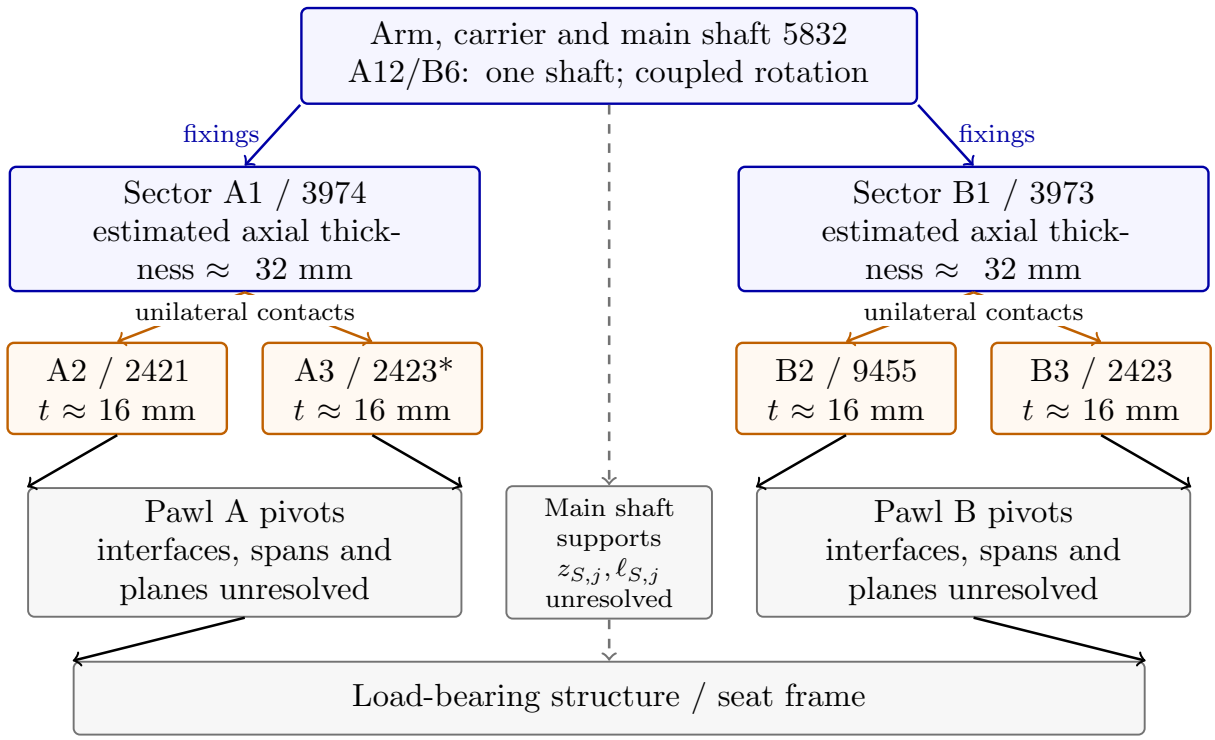


Figure 7: Functional topology. Connections represent load transfer or motion compatibility. Box positions do not represent axial order, scale, symmetry or simultaneous loading of every pawl. Supports are grouped by function; their physical number cannot be inferred from this graph.

Table 5: Parameters required by the axial model. Coordinates share one axial z direction; no numerical ordering is imposed.

Parameter	Meaning and current status
z_A, z_B	Planes of sector-resultant transfer to the carrier; separations unresolved.
$z_{P,k}$	Resultant loading plane of each pawl at its articulation; not automatically the sector mid-plane.
$z_{S,j}, \ell_{S,j}$	Position and effective length of each shaft or pivot support interface; number and dimensions unresolved.
e_k	Eccentricity between contact and effective support transfer; a parameter of spatial equilibrium.
k_j, c_k	Support stiffness and initial contact clearance; needed to determine load sharing and engagement under load.

and the actions from the arm and connections must be included. For a pawl, the relevant forces are its contacts, pivot reaction and, where modelled, spring and release loads. Internal forces cancel when the bodies are combined and must not be counted twice.

In this spatial frame, z is the mechanism rotation axis; it is distinct from the upward z axis used later in the seat-frame load model. For $\mathbf{F} = (F_x, F_y, 0)$, an axial offset e introduces

$$\mathbf{M}_\perp = (0, 0, e) \times (F_x, F_y, 0) = (-eF_y, eF_x, 0).$$

This retains eccentricities as parameters until the transfer planes are known. Two ideal 16 mm bands centred within a 32 mm sector would have centres at $e = \pm 8$ mm from the sector mid-plane. This is an illustrative arrangement, not a measurement or the lever arm automatically applicable to the fasteners.

A continuous shaft couples motion but does not impose instantaneous equal sharing. In the quasistatic opening-load scenarios, $M_A = \alpha M$ and $M_B = (1 - \alpha)M$ parameterise the allocation. Determining α requires clearances and stiffnesses, together with the contacts that can carry load at a common global angle. Independently analysed A–A and B–B poses cannot be combined as a demonstrated simultaneous pair. Four pawls likewise do not imply four equal forces.

7.4. Release and indexing

The identified A–A release actuation connects levers and links to withdraw the pawls. A13 is a rotation axis of that system and is retained to explain its motion. It may carry actuation reactions, although no opening-retention contribution is assigned to it. A9 is provisionally interpreted as a full-closure indexing pin, with no additional retention force credited in the current calculation.

The available axial reconstruction is functional and parametric. It connects components and defines their free bodies, but it does not replace a dimensioned axial section. Support positions, effective lengths and transfer planes would be particularly valuable for advancing from calculated contact forces to shaft bending, bushing pressure and pivot-pin bending.

7.5. A parametric transverse-support model

To make the support question quantitative without inventing dimensions, consider the *chosen* beam model in Figure 8, with two simple transverse supports at $z = 0$ and $z = L_S > 0$. Each support prevents transverse translation but supplies no bending couple. Transverse forces $\mathbf{F}_i = (F_{x,i}, F_{y,i}, 0)$ act at axial coordinates z_i, with any applied transverse couples $\mathbf{m}_{\perp,i}$. The diagram identifies the distances needed to calculate reactions; it is not an identified count or arrangement of the physical supports.

Define $\mathbf{Jv} = \mathbf{e}_z \times \mathbf{v}$ in the transverse plane and

$$\mathbf{Q} = \sum_i z_i \mathbf{J}\mathbf{F}_i + \sum_i \mathbf{m}_{\perp,i}.$$

Moment equilibrium about the first support and force equilibrium give

$$\boxed{\mathbf{R}_2 = \frac{\mathbf{J}\mathbf{Q}}{L_S}, \qquad \mathbf{R}_1 = -\sum_i \mathbf{F}_i - \mathbf{R}_2.} \tag{1}$$

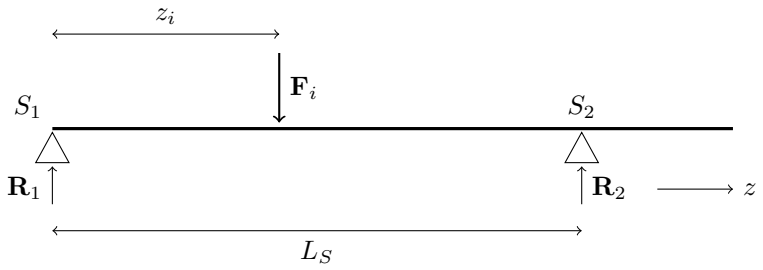


Figure 8: Chosen two-support reference model. Distances, support count and depicted force location are parameters, not a reconstructed axial stack.

Lengths in mm require couples in N mm; a couple tabulated in N m must first be multiplied by 1000. Only the transverse components enter this expression.

For one force $\mathbf{F}$ at $z = a$ and no couple,

$$\mathbf{R}_1 = -\left(1 - \frac{a}{L_S}\right)\mathbf{F}, \qquad \mathbf{R}_2 = -\frac{a}{L_S}\mathbf{F}.$$

A downward force between supports gives upward reactions. A load on an overhang can reverse one reaction; whether a physical interface can provide that reaction depends on its bilateral or unilateral constraints. The same equations accommodate carrier forces, pivot reactions or transported couples, provided they act on the selected free body in a common frame. Contact and support forces from different bodies must not be added twice.

Opening torque about z is distinct from these transverse bending couples. Its transfer must be resolved through arm, shaft, carrier and retaining branches; two transverse simple supports do not automatically react it. Axial thrust also requires separate constraints. No numerical shaft-bending diagram is reported because neither L_S nor the relevant transfer coordinates are established by the available images.

8. Contact, seating and opening restraint

8.1. From first contact to equilibrium

Locating the sector's first contact with a pawl held in its drawn orientation does not establish pawl equilibrium. A normal force whose line of action misses the pivot produces a moment; an ideal pivot cannot supply the counter-couple needed to balance it. The pawl must therefore be allowed to rotate towards a compatible contact configuration.

The contact model permits two motions: an additional clockwise sector opening δ and an anticlockwise pawl rotation ϕ, both relative to the source pose. Contact is followed along the original curve segments until an equilibrium seat is reached. Segment or arc membership, non-penetration of the complete outlines and normal orientation are checked. At an ideal corner, the admissible reaction may lie within the cone generated by the adjacent faces. The unilateral-equilibrium and force-transfer formulation follows general rigid-contact methods.[13]

Bodies are rigid, joints are ideal and contact is frictionless. The calculation represents the *incremental response to an opening load*: it does not determine the total force in the presence of

preload, weight and springs. These actions may control tracking during motion even when they are unnecessary for incremental equilibrium at the selected seat. Click dynamics and impact during clearance take-up have not been simulated.

Figures A.1 and A.2 show the nine calculated seated configurations and their admissible normal directions. The arrows in these figures act on the pawl, opposite to the sector normals defined below. Their resultant lines make the pivot-equilibrium condition visible, while the different normal arrangements distinguish an ideal corner from the B–B 32° case with separate contact points. These diagrams support the equilibrium argument; their arrows do not represent pressure footprints.

8.2. Sector and pawl equilibrium

For contact i, $\mathbf{C}_i$ is its position, $\mathbf{n}_i$ is the unit normal of the force *on the sector*, $N_i \geq 0$ is its magnitude and $\mathbf{P}$ is the pawl pivot. The signed moment arms are

$$h_i = \mathbf{C}_i \times \mathbf{n}_i, \qquad b_i = (\mathbf{C}_i - \mathbf{P}) \times (-\mathbf{n}_i). \tag{2}$$

Moment equilibrium of the sector about O and of the pawl about P requires

$$\sum_i h_i N_i = 1000M, \qquad \sum_i b_i N_i = 0, \qquad N_i \geq 0. \tag{3}$$

The factor 1000 converts N m into N mm. The pivot reaction balances the resultant contact force on the pawl without supplying an ideal reaction couple. The carrier connection balances the isolated sector. For two reaction generators,

$$\begin{bmatrix} h_1 & h_2 \\ b_1 & b_2 \end{bmatrix} \begin{bmatrix} N_1 \\ N_2 \end{bmatrix} = \begin{bmatrix} 1000M \\ 0 \end{bmatrix}.$$

An algebraic solution with $N_i < 0$ would require tension between the faces and is rejected. A normal outside the corner's admissible cone is also rejected. Two corner normals represent the admissible resultant of distributed contact in this idealisation; they do not represent two independent pawls or establish a contact area.

For a given seat, all incremental actions scale linearly with the assigned moment while the active contact set remains unchanged. This proportionality does not automatically extend to the opening of a contact, sliding, plasticity or loss of engagement.

8.3. Local kinematic condition

The normal gap g_i, measured in mm, cannot decrease from zero. With angular rates in radians per unit time and the adopted signs,

$$\dot{g}_i = -h_i \dot{\delta} + b_i \dot{\phi} \geq 0. \tag{4}$$

Setting $\dot{\delta} = 1$ gives the bounds on $\dot{\phi}$ required by all inequalities. If the lower and upper bounds are incompatible, no admissible infinitesimal opening motion exists in that model. All nine seats satisfy this condition and admit closing motion. This demonstrates *local, first-order* restraint of opening, rather than complete immobilisation of every degree of freedom or dynamic stability under finite disturbances.

Table 6: Resolved seated states. δ and ϕ are adjustments from the source pose. R/M is in N/(N m). The final column is the highest nominal normal stress among accepted sections at $M = 1000$ N m and $t = 16$ mm; it is not the component's local maximum.

Pose	Pawl	δ (°)	ϕ (°)	R/M	σ (MPa)
A-A 24°	front	1.197	0.571	13.061	112.6
A-A 29°	rear	0.943	0.669	12.129	44.5
A-A 33°	front	0.928	0.467	13.902	119.9
A-A 38°	rear	0.982	0.812	12.987	47.6
A-A 40°	rear	2.509	0.589	12.987	47.6
B-B 27°	front	1.048	0.475	12.823	119.8
B-B 32°	rear	0.549	0.618	11.040	45.2
B-B 36°	front	0.754	0.565	13.607	127.2
B-B 40°	rear	0.302	0.328	12.803	128.7

The resultant on the sector is $\mathbf{R} = \sum_i N_i \mathbf{n}_i$; its magnitude need not equal $N_1 + N_2$. A-A at 24° requires the rear pawl to be considered first, because it contacts the sector before the front pawl seats. A withdrawal trajectory allowing the front seat to be reached without penetration was verified. A-A at 38° and 40° reaches the same toothed seat; the larger adjustment in the final panel reflects its initial pose and is not a measured physical clearance.

8.4. Worked example: B–B at forty degrees

In the coordinate frame of this pose, the approximate positions are

$$\mathbf{C} = (79.929, 6.250) \text{ mm}, \qquad \mathbf{P} = (99.132, -54.646) \text{ mm}.$$

The two generators on the sector are $\mathbf{n}_1 = (-0.98996, -0.14135)$ and $\mathbf{n}_2 = (-0.23390, 0.97226)$. Their moment arms, in mm, are

$$(h_1, h_2) = (-5.111, 79.173), \qquad (b_1, b_2) = (-62.999, 4.427).$$

The solution per assigned N m is $N_1/M = 0.89167$ and $N_2/M = 12.68809$ N/(N m), with resultant $\mathbf{R}/M = (-3.85043, 12.21010)$ N/(N m). The smaller normal contributes an opposing moment on the sector, but is needed for pawl equilibrium. Dividing the applied moment by the radial distance OC would not give this result.

The seat is reached after $\delta \simeq 0.302°$ and $\phi \simeq 0.328°$ from the source pose. A later face-alignment candidate was rejected because it produced approximately 1.69 mm of penetration. This exclusion illustrates why a visually attractive normal can correspond to an impossible configuration.

8.5. The case with separate contact points

For B–B at 32°, two distinct points are retained: approximately $(90.531, 8.320)$ and $(86.342, -4.574)$ mm. Their normal magnitudes per N m are 11.03182 and 0.20592 N/(N m). The second is small, but contributes to equilibrium and must be retained when transferring moments to a section. An equivalent resultant may be transferred to an arbitrary point only if the corresponding transfer couple is included.

8.6. Equivalent point contact and the actual surface

Point forces provide resultants for rigid-body equilibrium. In a real component, force arises from a traction distribution over a surface. For an effective axial width b_{eff} and an in-profile contact length ℓ, the mean pressure over a hypothetical area is

$$p_{\text{mean}} = \frac{N}{b_{\text{eff}}\ell}. \tag{5}$$

With $N = 12.688$ kN and $b_{\text{eff}} = 16$ mm, choosing $\ell = 0.5$, 1 or 2 mm gives mean pressures of 1586, 793 or 397 MPa. This variation demonstrates the dependence on the selected area; none of these values is predicted as the actual pressure. Neither the full tooth height nor its 32 mm axial thickness automatically defines the load-bearing patch for a 16 mm pawl.

A local elastic solution would require radii, orientation, material properties and load per unit width. Hertzian models have specific curvature and contact-size assumptions; an ideal sharp corner does not supply the required radii.[16] Rounding, deformation and possible plastic zones would modify the contact patch and could also change the line of action. Global reaction forces and local pressure therefore remain connected problems with different input requirements.

9. Internal forces and conditional strength assessments

9.1. Nominal pawl sections

The discretised pawl geometry allows load transfer between contact and pivot to be examined. Three sections are sampled at 35, 50 and 65 % of the distance from the mean contact position to the pivot. Each section must intersect a single interval of material, retain the contacts on the isolated side and avoid an ambiguous representation of the pivot region. Of 27 candidate sections, 19 are accepted and eight excluded; the excluded cuts are not assigned a stress value.

Let w be the in-plane material width, t the axial thickness and $\mathbf{G}_s$ the section centroid. Contact actions on the isolated portion give an axial force N_{ax}, a shear force V and a section moment

$$M_s = -\sum_i (\mathbf{C}_i - \mathbf{G}_s) \times \mathbf{F}_{i,\text{pawl}}. \tag{6}$$

Here M_s is in N mm for forces in N and geometric coordinates in mm. For an ideal prismatic rectangular section,

$$A_s = tw, \qquad W_s = \frac{tw^2}{6}, \qquad \sigma_\pm = \frac{N_{\text{ax}}}{tw} \pm \frac{6M_s}{tw^2}. \tag{7}$$

The reported nominal normal stress is $\max(|\sigma_+|, |\sigma_-|)$. It excludes stress concentrations at radii, pivot bearing-pressure distributions and out-of-plane bending. The section equilibrium also gives shear force, but the reported normal stresses do not provide a three-dimensional equivalent-stress assessment.

For B–B at 40°, with $t = 16$ mm and an assigned moment $M = 1000$ N m, the highest value among accepted sections is 128.7 MPa. This value corresponds to the equilibrated contact configuration. Nor can the pawl with the largest tabulated stress be identified as the weakest actual component: accepted sections sample different regions and do not seek a global maximum.

For unchanged in-plane geometry, $\sigma \propto M/t$. If all in-plane dimensions change by a scale factor s_g, contact arms scale with s_g and the forces required for the same moment scale with $1/s_g$. Section

stresses scale with $1/s_g^2$ at fixed t, or $1/s_g^3$ if thickness also scales. Numerical precision of the coefficient does not replace dimensional calibration.

9.2. Two-fastener sector connection

The connection is idealised as a rigid sector with two fasteners of equal transverse stiffness, without frictional load transfer between mating faces. Let $\mathbf{Q}_j$ be the fastener positions, $\mathbf{G}$ the group centroid and $\boldsymbol{\rho}_j = \mathbf{Q}_j - \mathbf{G}$. The forces that the connection must exert on the sector are

$$\mathbf{F}_j = -\frac{\mathbf{R}}{2} + \frac{T_G}{\sum_k |\boldsymbol{\rho}_k|^2}\,\mathbf{J}\boldsymbol{\rho}_j, \qquad \mathbf{J}(x,y) = (-y,x), \tag{8}$$

where $T_G = -\sum_i (\mathbf{C}_i - \mathbf{G}) \times \mathbf{F}_{i,\text{sector}}$ is the couple required from the group. Thus, $\sum \mathbf{F}_j = -\mathbf{R}$, and the fastener moment cancels the contact moment. Direct and eccentric contributions are added as vectors before their magnitude is calculated. NASA RP-1228 provides the methodological reference for fastener groups and load-transfer mechanisms.[14]

Table 7: Transverse force magnitudes in the two fasteners per unit assigned moment, in N/(N m). Rigid model with two equivalent fasteners and no interface friction.

Seated state	Fastener 1	Fastener 2
A-A 24°	6.461	13.135
A-A 29°	10.596	6.842
A-A 33°	6.470	11.657
A-A 38°	10.965	6.407
A-A 40°	10.965	6.407
B-B 27°	12.476	6.623
B-B 32°	6.915	12.078
B-B 36°	11.083	6.665
B-B 40°	6.249	11.201

This response represents transverse demand in the selected connection model. A preloaded interface could transfer part of the load through friction; clearances, shank fits, face contact and housing stiffness would affect engagement. Thread stress is not calculated using $\pi 12^2/4$: the reported internal diameter does not identify the effective area of a nominal thread. Strength class, grip length, thread geometry, preload and shear-plane location are needed to compare these demands with fastener resistance.

9.3. Pins and bushings: bearing pressure, shear and bending

For a pawl resultant R, assumed diameter d_p and loaded width $t_p = 16$ mm, three estimates are distinguished:

$$p_{\text{proj}} = \frac{R}{d_p t_p}, \tag{9}$$

$$\tau_{\text{single}} = \frac{4R}{\pi d_p^2}, \qquad \tau_{\text{double}} = \frac{2R}{\pi d_p^2}, \tag{10}$$

$$M_{p,\max} = \frac{R(2L_p - t_p)}{8}, \qquad \sigma_{p,b} = \frac{32 M_{p,\max}}{\pi d_p^3}. \tag{11}$$

These expressions use forces in N and dimensions in mm, giving $M_{p,\max}$ in N mm and stresses in MPa; the example table converts the moment to N m. The first estimate is mean pressure over a projected area, not peak pressure around a bushing. The next two are average shear stresses in a solid pin with one or two ideal shear planes. The bending expression assumes two simple supports separated by L_p, with a centred uniform load over $t_p \le L_p$; it is not a verified model of the actual joint.

Table 8: Example for B–B 40°, $M = 1000$ N m, $R = 12.803$ kN and assumed $d_p = 24$ mm. Projected bearing pressure is 33.34 MPa; average single- and double-shear stresses are 28.30 and 14.15 MPa. Support spans are selected parameters.

L_p (mm)	$M_{p,\max}$ (N m)	$\sigma_{p,b}$ (MPa)
32	76.82	56.60
48	128.03	94.33
64	179.24	132.07

The example shows why a pin may require bending assessment as well as shear assessment. Support position and length govern this model. A bushing, shoulder, thread, hollow bore or eccentric load would require a different load-bearing section and distribution. The apparent identity of A3/B3 does not establish these joint properties.

9.4. Circular sections of the arm tube and main shaft

For an ideal circular annulus with outer diameter D and inner diameter d,

$$A = \frac{\pi}{4}(D^2 - d^2), \quad I = \frac{\pi}{64}(D^4 - d^4), \quad J = 2I, \quad W_b = \frac{2I}{D}, \quad W_t = \frac{2J}{D}. \tag{12}$$

The extreme bending and Saint-Venant torsional stress magnitudes are $\sigma_b = M_b/W_b$ and $\tau_t = T/W_t$, with moments in N mm. The relation between torque, section properties and deformation follows circular-shaft theory.[15]

Table 9: Ideal circular-section properties. A is in mm^2, I in mm^4 and section moduli in mm^3.

Component	D/d (mm)	A	I	W_b	W_t
Arm tube	60/50	863.9	329376	10979.2	21958.4
Ideal main shaft	60/32	2023.2	584701	19490.0	38980.0

The 60×5 arm tube and the 60/32 hatched shaft are different bodies. The clockwise opening moment acts about the mechanism axis; it is not automatically torque about the tube's longitudinal axis. Tube loading may predominantly involve bending, depending on the line of action. For a local state containing one normal and one shear stress component in an ideal ductile material, a possible criterion is $\sigma_{\mathrm{VM}} = \sqrt{\sigma^2 + 3\tau^2}$. Component maxima occurring at different locations must not be combined as though they acted simultaneously at one point.

The shaft section represents a smooth annular segment. It does not model shoulders, connections, holes, carrier contact or stress concentrations. Shaft torque $T(z)$ and bending moment $M_b(z)$ must be obtained from actions and supports in a common three-dimensional frame; section moduli alone do not determine them.

9.5. Materials, treatments and the meaning of capacity

No readable part-specific alloy or heat-treatment specification has been established for the pawls, sectors, main shaft or arm tube. Three core yield strengths, 355, 650 and 900 MPa, and tube values of 235, 355 and 460 MPa are retained as hypotheses. These levels explore sensitivity; they do not identify the B&M steel or establish verified lower and upper bounds. A machined appearance or visible bushing cannot establish alloy, carburising, quenching or hardness.

Manufacturer data provide context for such strength levels. SSAB lists structural tube grades with minimum yield strengths of 235, 355 and 460 MPa, among others.[26] Ovako's 42CrMo4 data distinguish product variants, delivery condition and section size, illustrating why an alloy name alone is insufficient to assign a component strength.[27] These are examples of documented steel products, without evidence that either supplier or grade was used in the restraint. Product form, heat treatment and surface condition would have to be established for the actual parts.

The reference elastic constants $E = 210$ GPa and $\nu = 0.30$ are consistent with the typical values listed in Ovako's 42CrMo4 data.[27] They are not calibrated properties or an alloy identification for these parts. Such constants could support a future compatibility analysis, but do not change the present statically determinate equilibrium. Surface treatment could influence contact, wear and fatigue without its hardness translating directly into the yield strength of the entire core.

For the cited B–B 40° section, dividing its 128.7 MPa nominal stress by each assumed yield strength f_y gives 0.363, 0.198 and 0.143 for these three cores. These are nominal stress/yield-strength ratios for one section and scenario. Inverting them and calling them restraint safety factors would omit other resistance modes and load uncertainty. A capacity conditional on that section criterion could be written as $M_y = f_y/c_\sigma$, but would remain a nominal submodel threshold rather than a component or assembly failure load.

No sufficiently validated tooth-root stress, fracture analysis, fatigue life or wear resistance has been obtained. These require local geometry and material properties, a load history and appropriate failure criteria. A subsequent finite-element analysis should retain the free-body diagrams, contact areas and provenance established here; mesh refinement cannot compensate for an incorrect support model.

10. From occupant dynamics to restraint loading

10.1. Why mass and an acceleration value are insufficient

The force received by the restraint is one of the occupant's contact actions. In an inertial frame, for the occupant's centre of mass G,

$$\mathbf{F}_{\text{seat}\to p} + \mathbf{F}_{\text{backrest}\to p} + \mathbf{F}_{\text{bar}\to p} + \mathbf{F}_{\text{other}\to p} + m\mathbf{g} = m\mathbf{a}_G. \tag{13}$$

By action and reaction,

$$\mathbf{F}_{p\to\text{bar}} = m(\mathbf{g} - \mathbf{a}_G) + \mathbf{F}_{\text{seat}\to p} + \mathbf{F}_{\text{backrest}\to p} + \mathbf{F}_{\text{other}\to p}. \tag{14}$$

Load distribution between contacts depends on posture, geometry, pad deformation and relative motion. Vehicle acceleration alone does not determine the acceleration of the occupant's centre of mass during a transition. A complete occupant model would also require angular equilibrium or dynamics.

Specific force is defined here as $\mathbf{f} = \mathbf{a}_G - \mathbf{g}$, making the first term in Equation (14) equal to $-m\mathbf{f}$. This separation avoids counting gravity twice when a reading already represents specific force. NASA's measurement-equation documentation provides the general accelerometer framework; it does not provide loads for this attraction.[17]

A train-mounted accelerometer measures specific force at its mounting point. Relating that record to the occupant's centre of mass also requires sensor orientation, rotational offset terms and relative body motion; the record alone does not determine the bar-contact force.[17]

As a one-dimensional illustration, a selected mass of 110 kg and $f_z = -0.7g$ would give approximately 755 N on the bar if that entire component were carried by it and no other contributions existed in that direction. A selected effective lever arm of 0.40 m would give approximately 302 N m. The example illustrates the required assumptions rather than establishing a service load.

10.2. Public operating context and the limits of comparison

Europa-Park's launch-era technical sheet for Silver Star reports approximately 20 seconds of airtime, with values reaching $-0.2g$.[24] This is historical information for one B&M Hyper, rather than an authenticated load envelope for Mako or the entire family. The duration is aggregate airtime, not a constant 20-second pulse at that value; the sheet gives no measurement protocol, seat position or filtering convention.

Schüttler and Wilhelm published accelerometer measurements on Silver Star. Their resultant-magnitude record contains approximately 20 seconds below $0.4g$; they selected that threshold because noise affects the resultant near zero.[25] A magnitude alone cannot recover the sign or direction of the load on the lap bar. These measurements support near-weightless intervals, but do not establish a signed peak force, an occupant-specific demand or restraint capacity.

These sources provide operating context for the force scale. A numerical service comparison still requires the signed specific-force components, their time alignment and the occupant/contact balance. A positive vertical peak carried mainly by the seat is not a downward restraint demand of the same magnitude. Selected load scenarios are therefore identified separately from reported ride data.

10.3. Opening moment and the restraint's own dynamic terms

The occupant-contact contribution to the moment about the axis is obtained by projecting $\mathbf{r}_C \times \mathbf{F}_{p\to\mathrm{bar}}$ onto that axis in the opening direction. The effective lever arm is the perpendicular distance to the line of action; it is not necessarily the full tube length.[18] The real arm balance also includes its weight, inertia, actions from other components and possible contacts. The following envelope applies selected forces to an equivalent geometric arm and excludes these additional terms.

Figure 9 defines the equivalent arm, contact position and normal/tangential force directions used in the moment reduction. The 250 mm segment represents the side-view extent of thigh contact, not a contact area or a transverse width between the legs. With non-negative normal pressure on this ideal segment, the centre of pressure lies between its endpoints. Its position cannot be established merely from the visible pad: a pressure distribution or occupant-contact model would be required.

10.4. What photographic rectification establishes

The photographic measurements described in Section 4 were taken from the second restraint counting from the foreground. The bore-based scale and tube-diameter check share the same

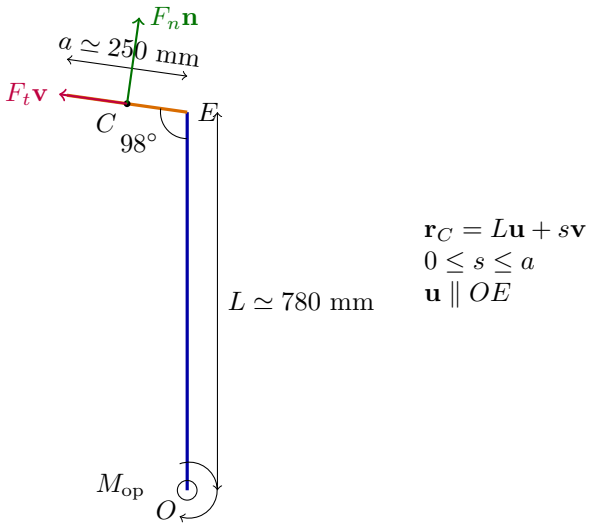


Figure 9: Coaxial reference geometry for the initial load envelope. The segment represents a contact direction; it does not reproduce the pad's curved profile, volume or actual surface offset. The indicated forces act on the bar.

reconstructed calibration, so a scale error would propagate in a correlated way through the measured lengths.

A homography relates points on a common plane; applying one does not automatically place surfaces at different depths in a common metric frame.[19] Complete rectification parameters and camera calibration are unavailable, and the pad surface is three-dimensional. The 780/250/98 geometry is therefore retained as a provisional estimate without a statistically quantified uncertainty. The public sources reviewed did not provide independent dimensions corroborating these three values.

11. From thigh contact to internal loads

The provisional photographic geometry uses a distance $L = 780$ mm from pivot O to junction E, an equivalent 250 mm contact segment and an interior angle $\beta = 98°$. The photographic calibration and its limitations are described in Section 4. The curved pad profile and out-of-plane offsets remain unresolved.

Let $s \in [0, 250]$ mm denote the centre-of-pressure position from E, F_n the force normal to the segment and F_t the tangential force positive towards its outer end. The selected normal produces clockwise opening. In a local frame with $\mathbf{u}$ parallel to OE,

$$\mathbf{r}_C = L\mathbf{u} + s\mathbf{v}, \qquad M_{\text{op}} = \frac{F_n[-L\cos\beta + s] - F_t L \sin\beta}{1000}.$$

Forces in N and lengths in mm give M in N m. Numerically,

$$M_{\text{op}} = \frac{F_n(108.555 + s) - 772.409F_t}{1000}.$$

For non-negative normal pressure on a straight segment, its centre of pressure lies between the endpoints. The central position $s = 125$ mm is an assumption; uniform pressure over the entire segment would be a sufficient condition. The interval is not a photographic error band.

Five normal forces, 500, 1000, 1500, 2000 and 3000 N; five positions, 0, 62.5, 125, 187.5 and 250 mm; and five ratios, $k = F_t/F_n = -0.2, -0.1, 0, 0.1, 0.2$, are selected. The ratios are directional scenarios, not measured friction coefficients. The resultant force magnitude is $F_n\sqrt{1+k^2}$.

The 125 loads are combined with five moment allocations, $\alpha = 0, 0.25, 0.5, 0.75, 1$:

$$M_{A-A} = \alpha M_{\mathrm{op}}, \qquad M_{B-B} = (1-\alpha)M_{\mathrm{op}}.$$

Each of the nine seated states is evaluated separately. These allocations establish neither simultaneous pose compatibility nor actual stiffness-based load sharing. A9 is provisionally interpreted as a full-closure indexing pin and receives no assigned retention contribution. Closing moments are not converted into opening moments by taking their absolute value: the unilateral opening response is left unevaluated.

For $F_n = 1000$ N, $s = 125$ mm and $F_t = 0$, the total moment is 233.555 N m; the selected 50 % allocation assigns 116.778 N m to the side under study. The table retains independent candidate states.

Table 10: Central reference case with 50 % of the moment assigned. Nominal stresses for 16 mm pawls; the highest value among accepted sections is reported for each case.

Seated state	Sector contact $\lvert\mathbf{R}\rvert$, kN	σ_{nom}, MPa
A-A 24°	1.53	13.2
A-A 29°	1.42	5.2
A-A 33°	1.62	14.0
A-A 38°	1.52	5.6
A-A 40°	1.52	5.6
B-B 27°	1.50	14.0
B-B 32°	1.29	5.3
B-B 36°	1.59	14.8
B-B 40°	1.50	15.0

For $F_n = 1000$ N, $s = 250$ mm and $k = -0.2$, the moment increases to 513.037 N m. With $s = 0$ and $k = +0.2$, it becomes a closing moment of -45.927 N m. Figure 10 shows this dependence on contact position and force direction, including the sign change between opening and closing. The largest opening moment in the grid is 1539.111 N m, at $F_n = 3000$ N, $s = 250$ mm and $k = -0.2$. This is a selected scenario extremum, not a capacity or measured demand. The grid does not cover arbitrary force directions.

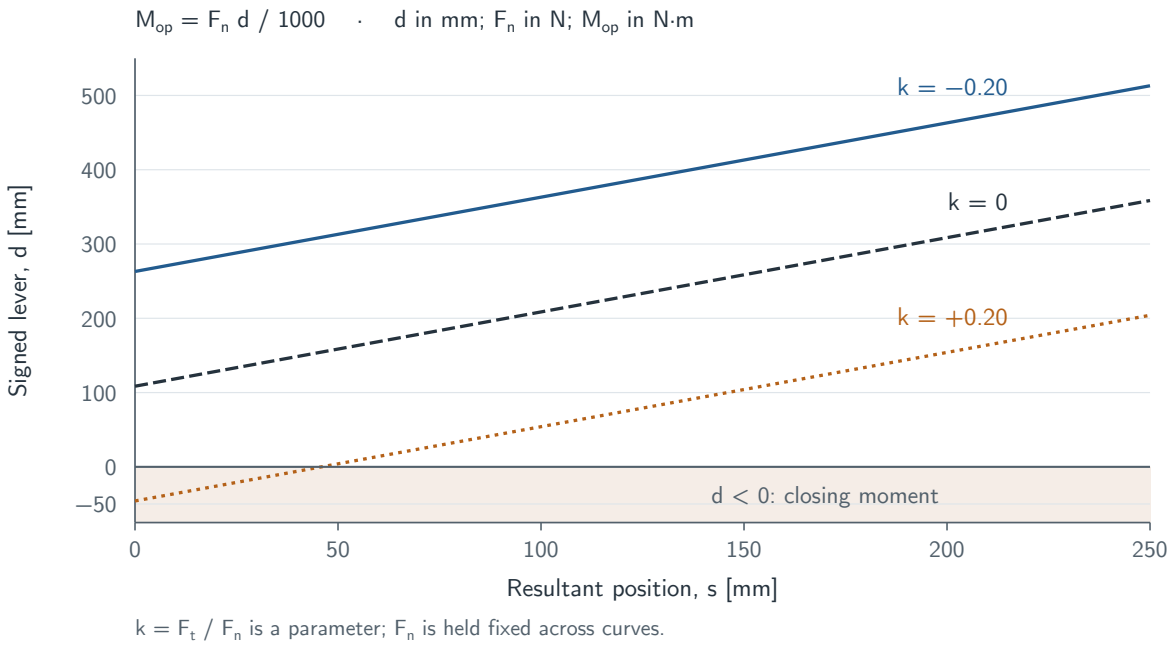


Figure 10: Signed effective lever arm per unit normal force for three selected directions. Negative values correspond to closing. The curves are scenarios for the ideal segment, not measurement uncertainty intervals.

11.1. Straight-beam reference model for the arm

The arm is represented by a $\varnothing 60 \times 5$ mm tube along OE, carrying the entire force, with coplanar loading and an ideal clamp at O. At a section q measured from O,

$$\begin{aligned} N &= 0.990268F_n + 0.139173F_t, \\ V &= 0.139173F_n - 0.990268F_t, \\ M(q) &= F_n[s + (780 - q)\sin 8^\circ] - F_t(780 - q)\cos 8^\circ. \end{aligned}$$

Here M is in N mm. Its largest absolute value occurs at an endpoint and $M(E) = sF_n$. With $A = 863.938$ mm^2 and $W = 10979.212$ mm^3,

$$\sigma_\pm = \frac{N}{A} \pm \frac{M}{W}.$$

Torque about the tube axis is zero in this planar idealisation. The side-to-side allocation does not divide the total force carried by this reference arm.

For $F_n = 1000$ N, $s = 125$ mm and $k = 0$, the extreme nominal normal stress is 22.419 MPa. With $k = +0.2$, the moment at O falls to 79.073 N m, while that at E remains 125 N m: the largest nominal stress is 12.564 MPa, compared with 8.381 MPa if only O is considered. Figure 11 compares the central-case linear stress responses of the full-force tube and the B–B pawl under two selected moment allocations. The slopes expose the different load-transfer assumptions, rather than ranking component capacities. The actual connection and supports have not been reconstructed.

Cross products, moment conservation, contacts, fasteners and stress components are checked separately. Of the 5625 responses, 4320 have positive opening moments, 1125 have zero incremental

load and 180 are not evaluated for opening because their moments tend to close the restraint. The 16875 section evaluations include geometrically inadmissible cuts, which are excluded rather than assigned zero stress. These results do not establish capacity, fatigue life or OEM material.

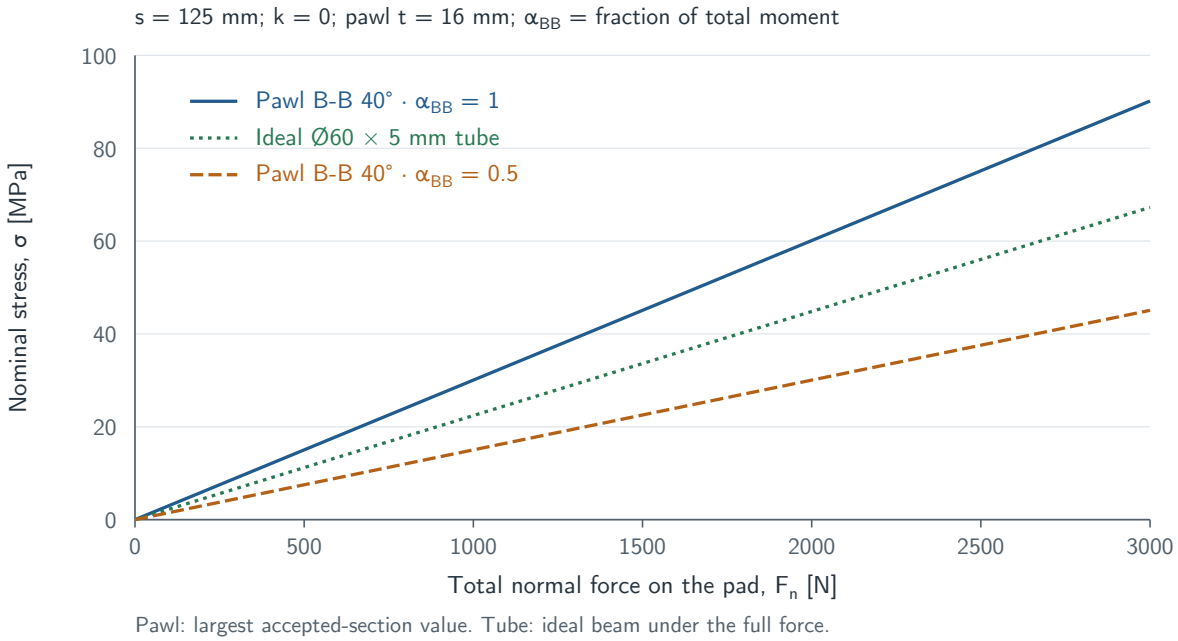


Figure 11: Central scenario, $s = 125$ mm and $k = 0$. Nominal response of the independent B–B 40° seated state for two moment allocations, and of the ideal straight tube carrying the full force. The slopes describe different submodels and do not compare ultimate capacities.

12. Conditional nominal load thresholds

The response coefficients can be inverted to address a more limited question than ultimate capacity: at what prescribed load does a selected nominal stress reach an assumed material yield strength? This section performs that inversion for the accepted pawl sections and an ideal straight tube. The resulting values are criterion crossings within those submodels. They do not establish fracture, release or loss of passenger retention.

12.1. Orientation recoverable from the CAD reconstruction

The tube walls in the combined drawing are not vertical. Their long edges point approximately 130.1° anticlockwise from CAD $+X$, or 40.1° to the left of CAD $+Y$. The two views repeat approximately the same geometry. Horizontal edges of the reconstructed housing/base provide a local mounting reference. Thus, an inclination relative to the normal of the drawn base can be recovered without a further photographic measurement.

The sequences preserve the orientation of each reconstructed pose; the figure layout translates and scales the panels without rotating them. Their final tube inclinations are approximately 40.1° in A–A and 39.7° in B–B, relative to CAD $+Y$. Small discrepancies between these values, the

nominal closing labels and the sector rotations are retained as reconstruction differences. The nominal 40° closing travel is not itself an exact absolute inclination relative to the seat.

For the reference calculation below, CAD $+X$ is assigned seat-forward and CAD $+Y$ upward. This is an explicit mounting hypothesis: the available material does not independently establish the relationship between that local base and the installed seat frame. A different mounting rotation would change the force lever arms.

The walls also reveal a distinction between the tube axis and the radius from the main pivot. Their outer separation is approximately 60 mm and their wall thickness approximately 5 mm, but the mean tube line misses pivot O by about 9.9 mm. The two reconstructed wall directions differ by about 0.06°; neither should be interpreted as an exact manufacturing line. The upper junction and the complete pad are absent from these truncated CAD contours.

12.2. An offset reference arm and directional loading

Let $\mathbf{u}_t = (\cos\phi, \sin\phi)$ point along the tube towards its upper end, with $\phi \simeq 130.1°$ in the chosen frame, and let $\mathbf{w} = (u_{t,z}, -u_{t,x})$ point to its right. Use a signed tube-axis offset $e \simeq -9.9$ mm and retain the provisional radial distance $L = 780$ mm. If the extended mean tube line contains the junction E, then

$$\mathbf{Q} = e\mathbf{w}, \qquad \ell = \sqrt{L^2 - e^2}, \qquad \mathbf{E} = \ell\mathbf{u}_t + e\mathbf{w}, \qquad \mathbf{u}_r = \frac{\mathbf{E}}{L}. \tag{15}$$

Here $\mathbf{Q}$ is the nearest point on the tube line to O. This constructs an unobserved upper junction from the stated assumptions; it does not measure that junction. The radius OE is approximately 40.9° to the left of the chosen upward direction, about 0.7° away from the tube direction.

The 98° angle in Figure 9 is interpreted between $E \to O$ and the equivalent contact segment. With $\beta = 98°$ and $\mathbf{J}(x, z) = (-z, x)$,

$$\mathbf{v} = -\cos\beta\,\mathbf{u}_r + \sin\beta\,\mathbf{J}\mathbf{u}_r, \qquad \mathbf{C} = \mathbf{E} + s\mathbf{v}, \qquad 0 \le s \le 250 \text{ mm}. \tag{16}$$

The angle between $O \to E$ and $\mathbf{v}$ is therefore 82°. The 98° value is not reapplied against the tube axis after introducing the offset. Figure 12 distinguishes the tube line from the radius OE and locates the forward/upward force components. That distinction determines the directional lever arms below.

For a force $\mathbf{F} = (F_x, F_z)$ applied to the bar, the clockwise opening moment about O is

$$M_{\text{op}} = \frac{C_z F_x - C_x F_z}{1000}. \tag{17}$$

Lengths are in mm and forces in N, giving N m. Pure upward and forward forces have lever coefficients $-C_x/1000$ and $C_z/1000$, respectively. The complete radius is not used as the lever for both directions. At the chosen $s = 125$ mm reference, these coefficients are approximately 0.615 and 0.522 m.

An upward–forward diagonal is represented by the unit vector $(1, 1)/\sqrt{2}$. A tabulated diagonal force is its resultant magnitude: each component is that magnitude divided by $\sqrt{2}$. These are prescribed structural load directions at C. Their transmission by an actual thigh-contact patch, including friction and the curved pad surface, has not been established.

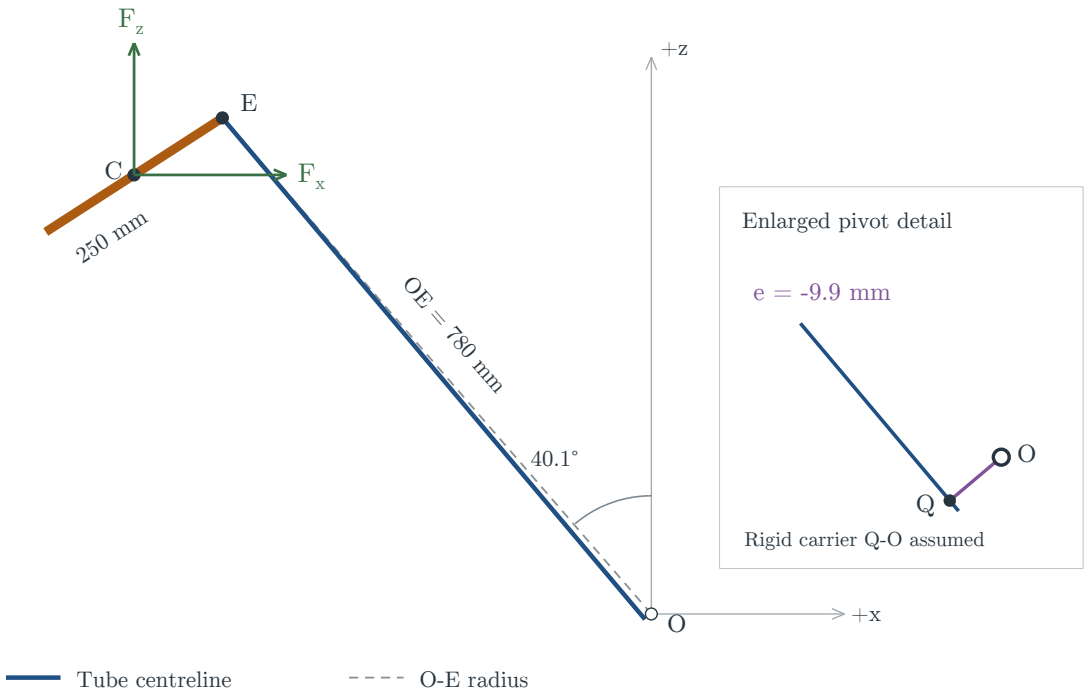


Figure 12: Offset reference arm and prescribed force components. The CAD axes are assigned forward and upward under the mounting hypothesis. The upper junction is constructed from the 780 mm radial distance and the measured tube-line direction/offset. The 40.1° annotation denotes the tube inclination; the radius OE is approximately 40.9° from upward. The pivot detail is enlarged; the connection between Q and O is an ideal rigid carrier.

12.3. Nominal criteria for the pawl sections

Let $c_{\sigma,i}$ denote the largest nominal normal-stress coefficient among the accepted sections of independent seated case i, at the assumed 16 mm thickness. For an assigned side moment,

$$\sigma_{\mathrm{nom},i} = c_{\sigma,i} M_i, \qquad M_{y,i} = \frac{f_{y,p}}{c_{\sigma,i}}. \tag{18}$$

The material strengths are the hypothetical core values introduced in Section 9. The inversion retains the same geometry, contact state and section coefficients; deformation or another mode could invalidate that extrapolation before the selected criterion is reached.

Table 11: Assigned moment at the chosen nominal normal-stress criterion among accepted pawl sections, in kN m. These are separate seated cases at $t = 16$ mm, not whole-part or assembly capacities.

	Assumed pawl core $f_{y,p}$ (MPa)		
Seated case	355	650	900
A–A 24°	3.15	5.77	7.99
A–A 29°	7.98	14.61	20.23
A–A 33°	2.96	5.42	7.51
A–A 38°	7.45	13.64	18.89
A–A 40°	7.45	13.64	18.89
B–B 27°	2.96	5.42	7.51
B–B 32°	7.86	14.39	19.92
B–B 36°	2.79	5.11	7.08
B–B 40°	2.76	5.05	6.99

For a chosen external unit direction $\mathbf{d}$, write $\mathbf{F} = F\mathbf{d}$ and $M_{\mathrm{op}} = hF$. If a fraction λ_i of that positive opening moment is assigned to case i, then

$$F_{y,p,i} = \frac{f_{y,p}}{c_{\sigma,i}\,\lambda_i\,h}, \qquad h > 0, \quad \lambda_i > 0. \tag{19}$$

The fraction refers to the particular side being evaluated: $\lambda_A = \alpha$ and $\lambda_B = 1 - \alpha$ when the earlier allocation convention is used. Closing moments remain outside this opening-pawl calculation. An unloaded branch has no criterion crossing from that assigned increment; this does not mean the complete restraint is unloaded.

The nine coefficients are evaluated independently using one chosen external reference geometry. This does not reconstruct the exact exterior orientation of every seated pose or establish that two independently solved seats can simultaneously carry their assigned fractions on the shared shaft. Their separate thresholds must not be minimised across arbitrary pose pairs and called an assembled capacity.

12.4. Nominal criterion for the ideal tube

The offset reference tube is clamped at $\mathbf{Q}$ and joined to O through an ideal rigid carrier. That carrier and its connection stresses are excluded. Along the straight segment from Q to E, axial

force is constant and the bending moment varies linearly. For unit direction $\mathbf{d}$, define

$$m_B = (C - B)_z d_x - (C - B)_x d_z, \qquad B \in \{Q, E\}, \tag{20}$$

$$c_t = \frac{|\mathbf{d}\cdot\mathbf{u}_t|}{A} + \frac{\max(|m_Q|, |m_E|)}{W}, \qquad F_{y,t} = \frac{f_{y,t}}{c_t}. \tag{21}$$

Here m_B is a lever in mm, $A = 863.938$ mm^2 and $W = 10979.212$ mm^3. Thus c_t is in MPa/N and the threshold force is in N. The sum is the extreme absolute axial-plus-bending normal stress per unit force; it is not a full local equivalent-stress or connection assessment. The entire prescribed force passes through this reference tube and is not divided by the side-allocation fraction.

The opening moment about O and the tube bending moment at Q differ by the moment transferred through the offset carrier. They must not be interchanged. The earlier coaxial reference beam remains the model used for its original load grid; the present construction is a distinct, more geometrically detailed scenario.

Table 12: Reference resultant forces at nominal normal-stress criteria, in kN, for $s = 125$ mm and the stated CAD-to-seat frame hypothesis. Both independent material assumptions are set to 355 MPa for this illustration; no common alloy is inferred. The diagonal has equal upward and forward components.

Criterion	Upward	Forward	Diagonal
Ideal tube	6.31	7.26	4.84
Sampled B–B 40°, $\lambda_i = 1$	4.48	5.28	3.43
Sampled B–B 40°, $\lambda_i = 0.5$	8.97	10.57	6.86

For example, assigning all opening moment to the sampled B–B 40° case gives a nominal upward-force criterion near 4.48 kN for that pawl hypothesis, while the tube hypothesis gives approximately 6.31 kN. Assigning half the moment to that pawl doubles its indicator to about 8.97 kN, while the tube result is unchanged. The comparison shows the consequence of a chosen allocation; it does not predict which member of the physical assembly yields first. The diagonal values likewise are not component forces or tested failure loads.

12.5. Chassis anchorage: observations and dimensional estimates

The mounting arrangement is interpreted from the photographs and assembly drawing as four chassis bolts, two on each side. The property class 8.8 was read from a photograph, and the drawing reference was transcribed as `N011-20`. These visual identifications do not constitute a material certificate or a complete joint specification. The suffix 20 is not interpreted as a nominal thread diameter.

The designation 8.8 encodes a nominal tensile strength of 800 MPa and a nominal yield-to-tensile-strength ratio of 0.8; nominal designation values must be distinguished from the applicable product and size-dependent minimum properties.[28] This class does not identify a unique alloy or establish the installed joint's resistance. ISO 898-1's published scope also excludes specification of shear resistance, fatigue resistance and torque/clamp-force performance.[29] No shear allowable or anchorage capacity is inferred from the marking alone.

Measurements on the approximately scaled drawing give about 25 mm between opposite parallel flats of the hexagonal head and about 15 mm for a feature described as the inner/thread diameter.

For comparison, a manufacturer's ISO 4017 catalogue lists M16 with a 24 mm head width across flats.[23] M16 is therefore retained as a plausible candidate, not an authenticated identification. The precise thread feature measured and the pitch remain unresolved; the 15 mm estimate is not adopted as a nominal diameter or an effective resisting area.

This chassis anchorage is a different connection from the two-fastener sector groups analysed earlier. The observations add information to the transmission chain, but the nominal thread size, bolt coordinates, axis directions, plate thicknesses, fits and preload remain unresolved. An eccentric force and moment would not, in general, produce four equal bolt forces. Face contact, friction and possible separation would also affect the load path. No numerical anchorage resistance is assigned here.

A section locating the bolt axes and plate interfaces would be useful for the assembled model. It could also clarify which housing/base direction corresponds to the seat frame. Such a targeted section would reduce a specific uncertainty without requiring a detailed reconstruction of every remaining component.

12.6. Failure modes and the meaning of the results

The parametric evaluation comprises 27 assigned-moment thresholds and 513 directional cases. The selected positions are $s = 0$, 125 and 250 mm; directions are upward, forward and their equal-component diagonal. Pawl fractions are 0.5 and 1, with assumed pawl strengths of 355, 650 and 900 MPa and tube strengths of 235, 355 and 460 MPa. Tables 11 and 12 summarise the moment criteria and one reference contact position, respectively. An independent calculation checked all 27 moment thresholds and 12 directional rows using polar geometry and section stresses reconstructed from primitive forces and widths. Force and moment transfer between O and Q was also checked.

Unassessed modes include tooth-root and local contact stresses, pivot and bushing behaviour, fastener and interface resistance, shaft/carrier/support stresses, out-of-plane actions and changes of contact with deformation. Nominal normal stress in sampled pawl sections is not the maximum stress in the entire pawl. The absence of those modes from a minimum does not make them stronger than the included ones.

Three events remain distinct: attainment of an assumed nominal yield criterion, fracture of a component, and loss of retention. Yield can precede continued retention, while disengagement could occur without yielding. Substituting ultimate tensile strength into an elastic normal-stress inversion would not establish either fracture or retention capacity. Post-yield redistribution and a compatible assembled contact model would be needed to follow the mechanism beyond the present criteria.

Converting one of these force indicators into a negative-g value would additionally require the occupant dynamics and contact allocation of Equation (14). Mass, specific force, other seat reactions, the contact position and the force direction must refer to the same scenario. The present values are conditional structural-load indicators, not attraction acceleration limits.

13. Assembly-level scenarios with a common arm rotation

13.1. From an assigned fraction to compatible rotations

The preceding force thresholds prescribe a moment fraction independently of deformation. A reduced assembly model can instead connect the arm to both locking branches and determine their

moment allocation from a common arm rotation. The calculation below does this for two selected rear-pawl seats. It adds an explicit compatibility hypothesis; it does not supply the missing strength assessments of the complete physical assembly.

Branches A and B denote the sides represented by sections A–A and B–B. The reference pair below uses their nominal final 40° configurations. Actual closure depends on the occupant and the resulting bar position; different riders need not use the same click or active pawl. This pair is a selected closure scenario, rather than a universal occupied position or a minimum dispatch requirement.

Rigid registration of the final A–A and B–B seated sectors to the combined-section reference gives clockwise offsets of approximately $g_A = 2.509°$ and $g_B = 0.726°$. Their difference is $\Delta g = 1.783°$, or 0.031114 rad. Each registered onset combines the rotation of its source pose into the common CAD frame with the local seating adjustment δ. For B–B 40°, $g_B = 0.4240° + 0.3021° = 0.7261°$; the atlas reports only the local δ. These are offsets in the reconstructed geometry, adopted as branch-onset scenarios. They are not measurements of physical backlash, elastic deflection or installed clearance.

13.2. An assumed torsional transmission network

Let θ be the common clockwise arm rotation from the reference geometry. Each branch contains an independent elastic transmission link of rotational stiffness K_i between the arm and a local sector, whose angle is ψ_i. The sector follows the unloaded arm until its unilateral rigid seat is reached. Further arm motion twists the assumed transmission link while the sector remains seated:

$$\psi_i = \min(\theta, g_i), \qquad M_i = K_i[\theta - g_i]_+, \qquad T = M_A + M_B, \tag{22}$$

where $[x]_+ = \max(x, 0)$ and angles in the constitutive law are in radians. Before the first seat, the model permits unloaded take-up; it has no quasistatic equilibrium for a nonzero opening force in that interval.

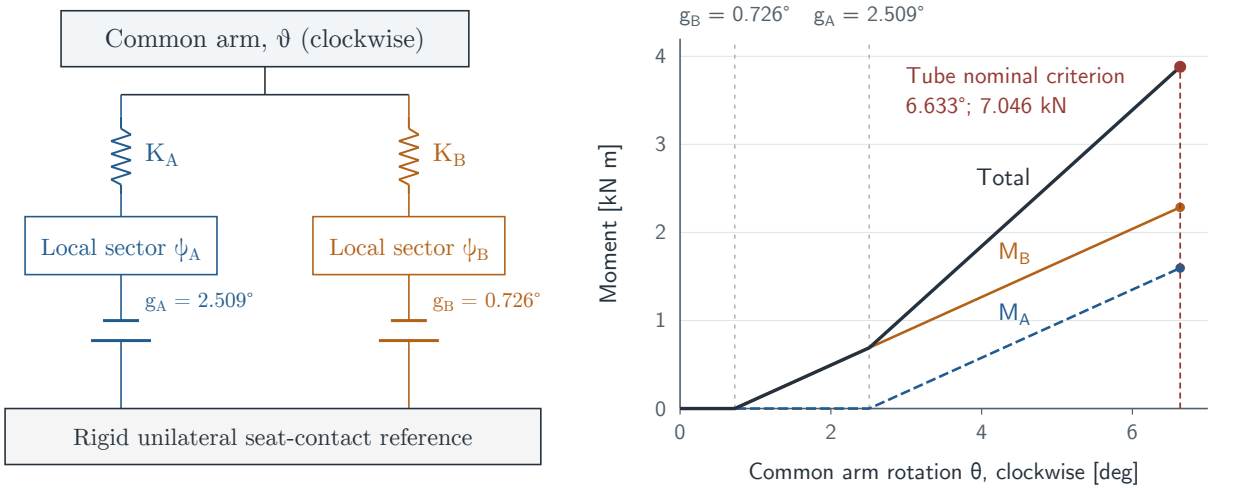
The calculation concerns incremental load. Bar weight, spring preload, inertia and the impact associated with taking up clearance are omitted. The unloaded take-up portion therefore does not predict the motion of a real restraint released into a gap. Figure 13 pairs the assumed branch connections in panel a with an illustrative piecewise moment–rotation path in panel b, using one of the parameter choices defined below.

This series-link construction preserves each local seated contact geometry: extra arm motion is assigned to the link, rather than to rotation of the sector through a stationary pawl. The links represent assumed arm-to-sector transmission compliance. They are not measured contact stiffnesses or independently verified models of the actual continuous shaft. Shared shaft segments could require a coupled stiffness matrix. Their strength is not assessed, and the rotational network does not introduce another grounded sector bearing or replace the sector fasteners and transverse load path.

13.3. External load and the first included criterion

The arm geometry rotates with θ, while the prescribed force direction $\mathbf{d}$ remains fixed in the assumed seat frame. With $\mathbf{R}_{\mathrm{cw}}(\theta)$ denoting clockwise rotation,

$$\mathbf{C}(\theta) = \mathbf{R}_{\mathrm{cw}}(\theta)\mathbf{C}(0), \qquad h(\theta) = \frac{C_z(\theta)d_x - C_x(\theta)d_z}{1000}, \qquad F(\theta) = \frac{T(\theta)}{h(\theta)}. \tag{23}$$



(a) Assumed torsional transmission topology. **(b)** Branch and total moments to the first included criterion.

Figure 13: Common-arm reference network and its conditional response. Each elastic link is in series with a rigid local seated branch; the actual shaft, carrier, transverse supports and link strength remain unresolved. The upward-force example uses the reconstructed onset order, $\eta = 0.25$, $\kappa = 1$, $s = 125$ mm and assumed pawl/tube yield strengths of 355 MPa. The curves end at the ideal-tube nominal criterion, at $\theta_* = 6.633°$ and $F_* = 7.046$ kN. This is an illustrative criterion crossing, not the physical assembly's capacity.

The quantities E, Q and the tube direction are rotated by the same transformation. Thus the full-force tube coefficient in Equation (21) is recalculated at each arm angle. The tube carries the entire applied force; its load is not divided between branches. Tube bending deflection does not feed back into this rigid external geometry.

For the selected final seats, the nominal pawl coefficients are $c_A = 0.047641$ and $c_B = 0.128689$ MPa/(N m). The included utilisation criteria are

$$U_A = \frac{c_A M_A}{f_{y,p}}, \qquad U_B = \frac{c_B M_B}{f_{y,p}}, \qquad U_t = \frac{c_t(\theta) F(\theta)}{f_{y,t}}. \tag{24}$$

The first included criterion occurs at the earliest angle θ_* where any of these reaches one; its force is $F_* = F(\theta_*)$. A pawl event can be located directly from $\theta_{y,i} = g_i + f_{y,p}/(c_i K_i)$; tube events are located along the load path. Positive lever arms and increasing force under load control are required. On each active phase,

$$\frac{dF}{d\theta} = \frac{T'h - Th'}{h^2} > 0. \tag{25}$$

The condition is checked through the evaluated paths. A chosen domain extends to 10° after first engagement. Reaching that boundary without a criterion is a censored calculation, rather than an unlimited resistance or an established elastic-validity limit.

13.4. Declared sensitivity parameters

The stiffnesses are varied through

$$\eta = \frac{K_B \Delta g}{T^{\mathrm{ref}}_{y,B}}, \qquad \kappa = \frac{K_A}{K_B}, \qquad T^{\mathrm{ref}}_{y,B} = \frac{355}{c_B} = 2758.59 \text{ N m}. \tag{26}$$

Both η and κ take the selected values 0.25, 1 and 4. The corresponding K_B values are 22.2, 88.7 and 354.6 kN m/rad. These are mathematical sensitivity choices, without physical calibration or probability weights. The fixed 355 MPa reference defines the stiffness scale even when other material strengths are selected.

Five onset families are evaluated. In degrees, the reconstructed pair is $(g_A, g_B) = (2.509, 0.726)$, the coincident pair is $(0.726, 0.726)$ and the reversed pair is $(0.726, 2.509)$. Separate A-only and B-only comparators retain each branch's reconstructed onset. The stiffness normalisation always uses the reconstructed $\Delta g = 1.783°$, including the coincident and reversed cases. The single-branch comparators are distinct ideal networks; they do not predict residual strength after a physical failure.

The grid uses $s = 0$, 125 and 250 mm; upward, forward and equal-component diagonal forces; assumed pawl strengths of 355, 650 and 900 MPa; and assumed tube strengths of 235, 355 and 460 MPa. Each onset family uses the three η and three κ values, except B-only, where κ is irrelevant and is counted once. These choices give 3159 scenarios.

13.5. Reference results and their physical meaning

Table 13: First included nominal criterion in the common-arm reference network: reconstructed onset order, $K_A = K_B$, $s = 125$ mm, $f_{y,p} = f_{y,t} = 355$ MPa. Forces are in kN; θ_* is the clockwise arm angle from the reconstructed reference. Material strengths and stiffnesses are assumptions.

	Upward force		Forward force		First criterion
η	F_*	θ_* (°)	F_*	θ_* (°)	
0.25	7.05	6.63	6.48	6.55	Tube
1	4.66	2.51	5.03	2.51	B pawl section
4	4.56	1.17	5.16	1.17	B pawl section

At $\eta = 4$, the B criterion is reached before A engages. At $\eta = 1$, it coincides with A's onset, where A still carries zero incremental moment. At $\eta = 0.25$, both branches carry load when the tube criterion is reached; their upward-case fractions are approximately 0.411 and 0.589. Equal stiffness therefore does not generally produce equal moment allocation when the onsets differ. The $\eta = 1$ upward and forward thresholds differ from the fixed-orientation B–B values in Table 12 because the arm has rotated by 2.509° when the B criterion is reached. Since A still carries no incremental moment at that event, the difference comes from the changed force lever arm, not an additional contribution from side A.

The soft-link case also requires large assumed rotations: approximately 4.12° in link A and 5.91° in link B at its upward criterion. The corresponding forward-case twists are 4.04° and 5.82°. For $\eta = 1$ and 4, the B twists are 1.78° and 0.446°, respectively, with A unloaded at the event. Reporting these rotations is essential: the stronger-looking force result in the soft case depends on uncalibrated transmission flexibility. Yielding or another limit in that transmission could invalidate the path before an included criterion is reached.

The complete evaluation contains 3159 scenarios: 2904 reach an included criterion and 255 reach the chosen angular boundary first. No evaluated path is excluded for a nonpositive lever or decreasing force. These counts describe the stated mathematical domain, not a population of

real restraints. The selected results do not define a guaranteed interval for actual capacity. Tooth and local contact stresses, pins, shaft and carrier resistance, transmission-link strength, sector connections, chassis anchorage, support behaviour and loss of engagement remain outside the included criteria.

13.6. Dependence on closure position

The initial position is not fixed by a universal click count. Occupant dimensions, posture and pad contact can stop closure at different angles. To examine this dependence without inventing an anthropometric model, the nine solved local cases are registered to the same angular reference. A–A 38° and 40° duplicate one seated stop, leaving eight distinct stops:

Nominal source configuration	Registered clockwise angle (°)
B–B 40	0.7261
A–A 38/40	2.5088
B–B 36	4.6212
A–A 33	7.5712
B–B 32	8.8930
A–A 29	11.4909
B–B 27	14.1956
A–A 24	17.2097

For a chosen initial arm angle θ_0, each side is assigned its next available reconstructed stop,

$$g_i(\theta_0) = \min\{g : \ g \text{ is a known stop on side } i, \ g \geq \theta_0\}. \tag{27}$$

This is an explicit stop-selection hypothesis. Ordering independently solved seats does not establish the actual pawl tracking, spring response or three-dimensional path during closure. The A–A 24° seat, in particular, depends on the conditional rear-pawl withdrawal discussed in the contact analysis. These pairs are conditional model configurations, rather than identified states of particular occupants.

Seven intervals contain a known next stop on both sides. Table 14 selects each interval's midpoint to avoid boundary ambiguity; at an exact boundary the $g \geq \theta_0$ rule retains the stop at that angle. The familiar nominal labels are not obtained by subtracting θ_0 from 40°. No minimum number of clicks or dispatch requirement is inferred.

The initial angle selects a pair and its unloaded take-up. Geometry remains rotated by the absolute θ, rather than being reset to $\theta - \theta_0$; positions within one interval therefore share the same subsequent loaded path in this idealisation. The stiffness normalisation is unchanged across all pairs. Reusing these stiffnesses for different front and rear pawls is an additional assumption, not a finding that their transmission compliances are equal.

Beyond the last known B stop at 14.1956°, the reconstruction contains no next B stop to select. That region is left unevaluated; it is not treated as an A-only restraint or assigned zero B resistance. Positions beyond the last A stop are likewise outside the available coverage.

The closure-position evaluation adds 5103 scenarios: seven pairs with the three η, three κ, three contact positions, three force directions and independent three-value pawl and tube strength

Table 14: Closure-position scenarios with $K_A = K_B = 88.7$ kN m/rad, $s = 125$ mm and assumed pawl/tube strengths of 355 MPa. Angles are clockwise from the reconstruction reference, not installed seat angles. Forces are in kN. Parentheses identify the first included criterion: A or B for the sampled pawl section; T for the ideal tube.

Initial θ_0 interval (°)	A seat	B seat	Upward	Forward
$[0, 0.7261]$	38/40	40	4.66 (B)	5.03 (B)
$(0.7261, 2.5088]$	38/40	36	6.82 (T)	6.66 (T)
$(2.5088, 4.6212]$	33	36	5.05 (B)	4.75 (B)
$(4.6212, 7.5712]$	33	32	7.45 (A)	6.21 (T)
$(7.5712, 8.8930]$	29	32	7.76 (T)	6.05 (T)
$(8.8930, 11.4909]$	29	27	8.24 (T)	5.86 (T)
$(11.4909, 14.1956]$	24	27	6.64 (B)	4.41 (B)

sets. Of these, 4923 reach an included criterion and 180 reach the chosen angular boundary first. Different pairs can change the first-loaded side, active pawl, stress coefficient and external lever. Because the stress coefficients sample different regions, these comparisons do not rank the actual strength of the restraint by click position.

13.7. Load scale and rider demand

With $g_0 = 9.81$ m/s^2, assigning the complete vertical specific-force component of a selected 150 kg occupant to the bar gives 0.294 kN at $f_z = -0.2g_0$ and 1.472 kN at $f_z = -g_0$, both below the 4.66 kN upward reference criterion. These are prescribed demand cases, not measured operating peaks or a verified fit assessment. SeaWorld's accessibility guide conditions riding on proper restraint fit and body-weight distribution, and lists Mako among rides that may not accommodate larger guests.[12] A maximum admissible body mass cannot therefore be inferred from closure alone without body dimensions and seat/pad geometry. The comparison tests the model's force scale; it does not demonstrate service adequacy for the heaviest guest who can fit.

13.8. Illustrative occupant-load checks

A rider mass is not itself a restraint load or a passenger limit. To make the distinction quantitative, select a specific-force vector $\mathbf{f} = (-k_x g_0, -k_z g_0)$ in the assumed forward/upward frame, with $g_0 = 9.81$ m/s^2. If its entire opposite contact component is assigned to the bar and all other occupant contacts are omitted,

$$\mathbf{F}_{\text{req}} = mg_0(k_x, k_z), \qquad F_{\text{req}} = mg_0\sqrt{k_x^2 + k_z^2}. \tag{28}$$

Gravity is already accounted for in specific force; another mg_0 is not added to this expression. The negative longitudinal component represents a selected braking scenario, giving a forward force on the bar. This construction is a prescribed demand, without demonstrating its actual transmission by the pad.

The historical Silver Star figure provides context for the $0.2g$ scale, subject to the convention and source limitations discussed in Section 10. The values $k_z = 0.5$, 1 and 2, and $k_x = 0$, 0.5 and 1, are selected sensitivity cases; they are not measured B&M peaks or a design envelope.

Table 15: Illustrative upward bar-force demand, in kN, if the complete selected vertical specific-force component is carried by the bar. Masses are scenarios, not rider limits. The values do not include gap-impact dynamics.

Selected f_z/g_0	100 kg	125 kg	150 kg
−0.2	0.196	0.245	0.294
−0.5	0.491	0.613	0.736
−1.0	0.981	1.226	1.472
−2.0	1.962	2.453	2.943

Their simultaneous occurrence is also hypothetical. Masses of 100, 125 and 150 kg are combined independently with all seven closure pairs, without assigning a particular bar position to a body mass.

The 36 demand vectors are compared with the first included criterion along their own normalised force direction in each of the seven closure networks, giving 252 comparisons. The reference uses $K_A = K_B = 88.7$ kN m/rad, $s = 125$ mm and assumed pawl/tube strengths of 355 MPa. All selected demands lie below an included criterion in that declared calculation. For example, the final-seat pair gives a 4.66 kN upward criterion, compared with 0.294 kN for the selected 150 kg, $k_z = 0.2$ case or 1.472 kN for $k_z = 1$.

The more severe selected combination $m = 150$ kg, $k_z = 2$, $k_x = 1$ gives a 3.290 kN resultant force on the bar. The final-seat network reaches its first assumed nominal yield stress at a force of 3.561 kN along that direction. Dividing the applied force by this threshold gives $3.290/3.561 \simeq 0.924$, the largest such force/threshold ratio among these 252 comparisons. Thus the prescribed force is approximately 92.4% of that model threshold force. This is not a physical safety factor: the assumed material, contact position, stiffness, closure path and omitted resistance modes remain decisive.

For these selected static scenarios, every prescribed demand vector remains below the first included nominal criterion of its declared network. This comparison neither bounds actual service demand nor establishes a verified operating margin for any rider mass. Signed load histories at the relevant seat, occupant motion, contact redistribution and gap-impact dynamics would be needed for that comparison; the strength of the complete transmission and anchorage would also need assessment.

14. Numerical verification and model sensitivity

14.1. Consistency checks

Each seated configuration is checked against the source outlines and its free-body diagram. Checks include contact membership on CAD segments, profile non-penetration, non-negative normal magnitudes, admissible normal cones and moment equilibrium of the sector and pawl. The kinematic inequalities are also checked for local restraint of opening. Moment residuals for the nine solutions are of order 10^{-13} N mm per assigned N m, measuring arithmetic closure rather than drawing accuracy.

A total of 90 rigid coordinate transformations were used to check scalar invariance and vector transformation. For B–B at 32°, each contact is retained when transferring moments to sections.

Fastener-group forces balance both the resultant and the moment about the group centroid. A solution is not retained merely because its force magnitude appears plausible.

The load envelope uses coefficients obtained from the external geometry and nine seated configurations. Load cross products, conservation of $M_A + M_B = M_{\mathrm{op}}$, stress components and tube results at five sections per load are checked. A separate implementation recalculated forces and moments row by row from these inputs; the largest force residual was of order 4×10^{-12} N. These checks do not identify missing material or support properties.

14.2. Interpreting counts and exclusions

The 125 external loads produce 625 moment assignments and 5625 independent seated-state responses. Of these responses, 4320 have a positive opening moment, 1125 represent zero incremental load and 180 are not evaluated for opening because their moment tends to close the restraint. The 16875 section evaluations include geometrically inadmissible cuts, which are excluded rather than assigned zero stress. Zero responses denote no increment from the selected load, not absence of preload in the real mechanism.

The grid uses discrete positions and directions. Its extrema can be verified within that selected domain, but are not universal bounds for every force direction. Even in the plane, a resultant with unrestricted orientation can have a lever arm equal to the radius $|\mathbf{r}_C|$, exceeding the values obtained within the selected tangential-ratio range. Defining the domain is as important as reporting its maximum.

14.3. Priorities for uncertainty reduction

Four dependencies are distinguished. In-plane scale changes moment arms and section dimensions; axial thickness changes load-bearing areas; seating geometry and radii control the line of action; and external load depends on position, direction and occupant interaction. Normalisation per N m separates the geometric response from the applied-load scenario, but does not remove geometric uncertainty.

Statistical distributions are not assigned to unmeasured errors. Variations in s, k, α, yield strength or pin span are deterministic scenarios. Future probabilistic propagation would require justification of input distributions and correlations. Similarly, increasing grid density does not, by itself, improve the physical reliability of its assumptions.

15. Published ride measurements and conditional demand histories

The preceding calculations prescribe forces at the pad. An experimental ride record can improve the choice of those forces, but it does not measure them directly. This extension examines published B&M Hyper measurements, recovers numerical readings from a Mako video, and compares explicit contact hypotheses with the existing structural model. The resistance models and their dimensional provenance remain those of Sections 12 and 13. The new observations are not used to tune an assumed material strength or transmission stiffness to produce a favourable result.

15.1. Experimental evidence and source quality

Table 16 distinguishes instrumented records from operating descriptions and educational illustrations. The public search covered the named rides, academic publications, park documents, instrument reports, video descriptions and first-person measurement accounts. No public numerical Mako acquisition file was located in that search. This is a statement about the material located, not proof that no such file exists.

Table 16: Measurement evidence relevant to the extension. Unknown settings remain unknown; a device capability is not an acquisition setting.

Source	Measurement or claim	Quantitative use here
Mako, CoasterStats (2025)	Chest-mounted three-component overlay; professional equipment and park cooperation stated. Model, sampling, exact filtering and calibration not established.	Same-frame numerical display readings and source-frame timing; no measured pad force.[31]
Silver Star, Schüttler (2011)	Four front/rear rides; vest-mounted PASCO and PHYWE loggers. Their hardware supports up to 100 Hz; actual run rate unspecified.	Acquisition context and signed-axis caution; original files listed on a companion DVD, not recovered as a public download.[32]
Silver Star, Schüttler and Wilhelm (2011)	About $3.6g_0$ maximum norm and 20 s cumulative norm below $0.4g_0$.	Near-weightlessness context; the norm gives no signed vertical minimum. Same experimental lineage as the thesis.[25]
Shambhala, 0Kajuna0 (2017)	Mouth-mounted Hero5 Session; author reports 200 Hz acceleration, about 400 Hz gyro and averaging for the video.	Methods and qualitative context. No calibrated force history or validated extrema adopted.[35]
Silver Star, historical park sheet	Park-reported $-0.2g_0$ and about 20 s aggregate airtime; no measurement protocol.	Historical context only; indexed primary text recovered, direct retrieval unavailable.[24]
Apollo's Chariot and Nitro workbooks	Educational graphs or suggested factors, without an acquired numerical series.	Excluded from demand validation. Apollo's workbook expressly disclaims scientifically accurate ride testing.[36, 37]

The Silver Star thesis is especially useful for avoiding a sign error: its PHYWE component plot rests near $-1g_0$ vertically, whereas the Mako overlay rests near $+1g_0$. Compression has opposite displayed signs. The thesis component and norm figures use separate instruments and clocks; they

cannot be merged into a simultaneous vector.[32] No verified downloadable, calibrated, synchronised history was established for Diamondback or Nitro during this search. Values from an educational exercise are not substituted for missing measurements.

CoasterStats' general methods accounts describe calibrated equipment, filtering and mounting that varies with permission. An earlier team account mentions a 5 Hz Butterworth filter and offers an unprocessed record for research on request. Those dated statements do not establish the settings, present availability or uncertainty of the 2025 Mako acquisition.[33, 34] Similarly, the Shambhala report's head-mounted measurement is not a seat-fixed record; the author's processing description and unsmoothed chart are retained as separate evidence.[35]

15.2. Signal convention and the measurement location

Write $g_0 = 9.81\ \mathrm{m\,s^{-2}}$ and $\boldsymbol{f} = \boldsymbol{a} - \boldsymbol{g}$. In the assumed seat frame, x points forwards, y left and z upwards; define the dimensionless components $q_x = f_x/g_0$, $q_y = f_y/g_0$ and $q_z = f_z/g_0$. A stationary upright accelerometer showing approximately $q_z = +1$ is consistent with specific force, not gravity-subtracted coordinate acceleration. The Mako trace has this baseline; the lift also redistributes the approximately unit vector between axes. This supports that interpretation of the displayed signal. Adding another body-weight term to $m\boldsymbol{f}$ would count gravity twice.

The recording is identified as chest-mounted.[31] Treating its components as the occupant centre-of-mass specific force, aligned with the assumed seat axes, is an additional hypothesis in the calculations below. For a moving point S relative to a vehicle reference O, rigid-frame kinematics give

$$\boldsymbol{a}_S = \boldsymbol{a}_O + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r}) + 2\boldsymbol{\omega} \times \boldsymbol{v}_{\mathrm{rel}} + \boldsymbol{a}_{\mathrm{rel}}. \tag{29}$$

Here $\boldsymbol{r}$ is the position of S relative to O, $\boldsymbol{\omega}$ is the reference frame's angular velocity, and $\boldsymbol{v}_{\mathrm{rel}}$ and $\boldsymbol{a}_{\mathrm{rel}}$ are the relative velocity and acceleration measured in that rotating frame. The relative terms vanish only for a point fixed to that frame. A chest sensor can include torso motion and rotation relative to the seat. Angular-rate data could assist frame transformation and rigid-point corrections, but would not by themselves determine occupant–pad contact forces or remove unknown torso motion. A properly aligned, seat-fixed accelerometer can still provide useful local specific force without a gyroscope.[17]

Large positive vertical values are principally associated with support from the seat in the usual seated contact configuration. They are not an equal force trying to open the bar. Conversely, positive q_z does not require zero bar force: opposing seat and pad reactions can be substantial while their vertical difference remains mg_0q_z.

15.3. Recovering numerical readings from the Mako video

The overview graph superposes the longitudinal, lateral and vertical traces. A colour-mask extraction retained 1900 graph columns, but only 1457 had both longitudinal and vertical colour support. Stroke width and overdraw also made some column medians poor estimates of short peaks. The numerical overlay in the video provides a more direct route: its four printed values remain readable when one graph trace covers another.[31] The overview is therefore used as a source cross-check rather than the principal numerical input below.

Readings were extracted from a 1920×1080, 60 frames/s copy of the video. A regular selection at 20 display frames/s was supplemented by all frames in five short peak windows, totalling

129 additional inspected frames before duplicate removal. Classification of the fixed printed font, corroborating text recognition and manual image checks distinguish confirmed readings from rejected or ambiguous characters. Minus signs require particular care on bright backgrounds. The checks include 181 manually read values in the regular selection, 120 targeted sign checks and 44 values in the peak windows; these sets overlap and do not constitute independent experiments or a statistical error estimate.

After merging duplicate source frames, the record contains 2898 selected instants. Confirmed longitudinal, lateral and vertical readings are available at 2799, 2687 and 2773 of them, respectively; 2681 retain both longitudinal and vertical values and 2513 retain all three components. These are fractions of the selected frames, with extra sampling near some peaks, rather than fractions of ride duration. For the structural comparisons, every required component must be confirmed in that same frame. An interpolated value cannot supply a missing component.

The printed norm provides a transcription check,

$$q_{\mathrm{norm}} \simeq \sqrt{q_x^2 + q_y^2 + q_z^2}. \tag{30}$$

Two-decimal rendering permits small residuals. Agreement does not validate component signs, instrument calibration, mounting or the underlying processing. The displayed decimal resolution is not a measurement-uncertainty bound. The resulting observations remain processed, rounded display readings, not recovered original accelerometer samples.

Table 17: Confirmed numerical-display extrema and the overview's written extrema, in units of g_0. Recovered extrema use confirmed readings only. The final onride longitudinal minimum indicator reads -0.85, in conflict with the overview's -1.50.

	Recovered display		Written overview	
Quantity	Minimum	Maximum	Minimum	Maximum
Longitudinal q_x	-0.85	$+0.84$	-1.50	$+0.84$
Lateral q_y	-0.71	$+0.79$	-0.71	$+0.79$
Vertical q_z	-0.84	$+3.72$	-0.85	$+3.72$
Scalar norm	0.03	3.73	0.02	3.74

Table 17 separates the recovered display values from the overview's written extrema. Native-frame checks recover the reported lateral extrema and the longitudinal maximum. A substantive longitudinal discrepancy remains: the overview reports $-1.50g_0$, whereas the final onride longitudinal minimum indicator reads $-0.85g_0$, matching the lowest recovered numerical longitudinal reading. The reconstruction preserves this disagreement and does not insert a $-1.50g_0$ event. The $0.01g_0$ differences in the vertical and scalar extrema likewise remain explicit. Sampled, rounded displays need not reproduce extrema calculated from a finer or differently processed record; the source does not establish the cause of each difference.

15.4. Video-frame timing and visual completion

Each recovered value is attached to the presentation timestamp of its identified source video frame. The selected interval is 7.767–148.317 s on that container clock. Pixel-identical comparisons at 43 regular/native-frame matches establish the phase of the 20 frames/s selection. Container

timestamps are stored to the millisecond; that storage precision does not establish accelerometer timing accuracy or sensor/video synchronisation. The five native peak windows make the final observation grid intentionally nonuniform.

This frame identification removes the need to infer a time axis from the attraction's duration or to use an affine registration of the overview. It supports locating the recovered observations within the video. The video's 60 frames/s and the 20 frames/s extraction interval still do not establish the sensor acquisition rate, filtering or effective measurement bandwidth.

Figure 14 presents continuous curves for readability. Confirmed readings are opaque; pale segments linearly join the remaining illegible intervals. During the active portion before 125 s, the largest gaps between confirmed anchors in the regular selection are approximately 0.30 s longitudinally and vertically, and 0.35 s laterally. Larger contrast gaps near the station reach 0.80 s between anchors for the horizontal components. Every completed value retains an estimated status. This visual completion cannot rule out a hidden peak, and none of its interpolated values enters the demand maxima or structural comparisons. Jerk, exact acquisition-level pulse duration, fatigue spectra and gap-impact velocity are not inferred.

15.5. Three distinct occupant-contact hypotheses

Let (B, U) denote the forward and upward force *on the bar*. Its reaction on the occupant is $(-B, -U)$. The following hypotheses are evaluated separately; their curves must not be combined into one purported measured bar-force history.

Upward force only.. For confirmed $q_z < 0$, assign $U = -mg_0 q_z$ and $B = 0$. This is a one-component force comparison. Longitudinal and lateral demands are left to unspecified other contacts; no complete occupant-contact solution is claimed.

Upward and forward forces assigned to the bar.. For frames with confirmed $q_x < 0$ and $q_z < 0$, prescribe

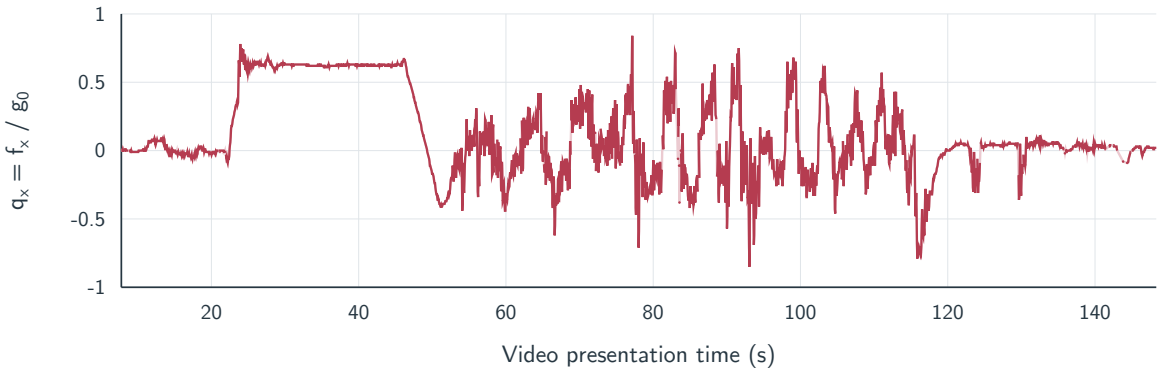
$$B = -mg_0 q_x, \qquad U = -mg_0 q_z. \tag{31}$$

This balances the chosen in-plane specific-force components if the bar supplies them. It does not demonstrate that the curved pad can transmit that resultant at one frictionless normal contact. Lateral loading and occupant moment balance remain outside this planar construction.

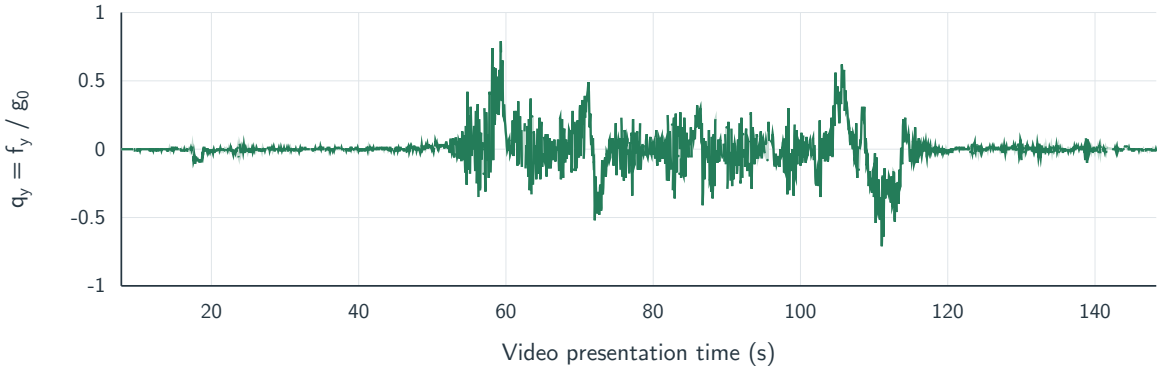
Seat and pad active during braking.. Choose a frictionless horizontal seat, with upward reaction N, and a single compressive pad force. Let α_{F} be the direction of the force on the bar, measured above the forward horizontal in the assumed seat frame. This force angle is distinct from both the restraint's closing angle and the earlier moment-sharing fraction α. Assign all longitudinal braking to the pad. Translational equilibrium gives

$$B = -mg_0 q_x, \qquad U = B \tan \alpha_{\mathrm{F}}, \qquad N = mg_0(q_z - q_x \tan \alpha_{\mathrm{F}}), \qquad F_{\mathrm{bar}} = B \sec \alpha_{\mathrm{F}}. \tag{32}$$

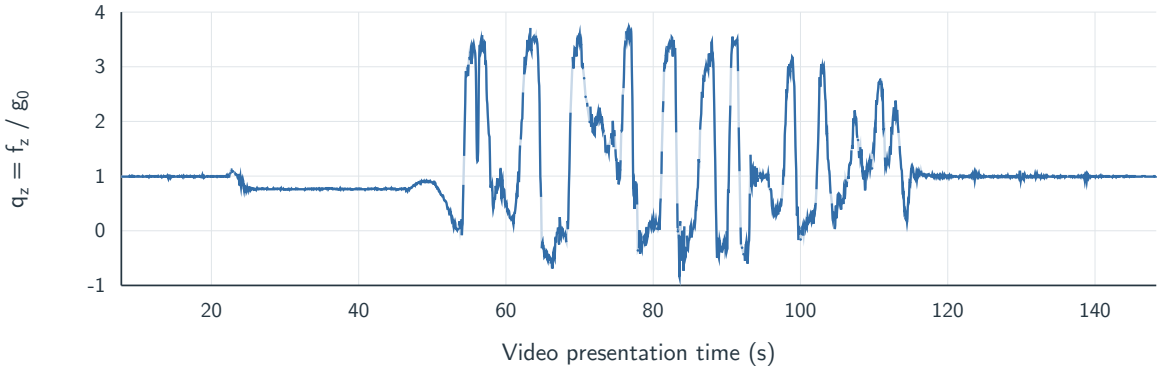
Only $q_x < 0$ and $N \geq 0$ are retained. A negative seat reaction invalidates this active-contact configuration; it is rejected rather than clamped to zero. Removing the seat would require a different contact solution. Three seated-braking cases use $\alpha_{\mathrm{F}} = 30°$, $45°$ and $60°$. They prescribe hypothetical force directions rather than measured pad normals or the $98°$ arm geometry. The



(a) Longitudinal, positive forwards.



(b) Lateral, positive left.

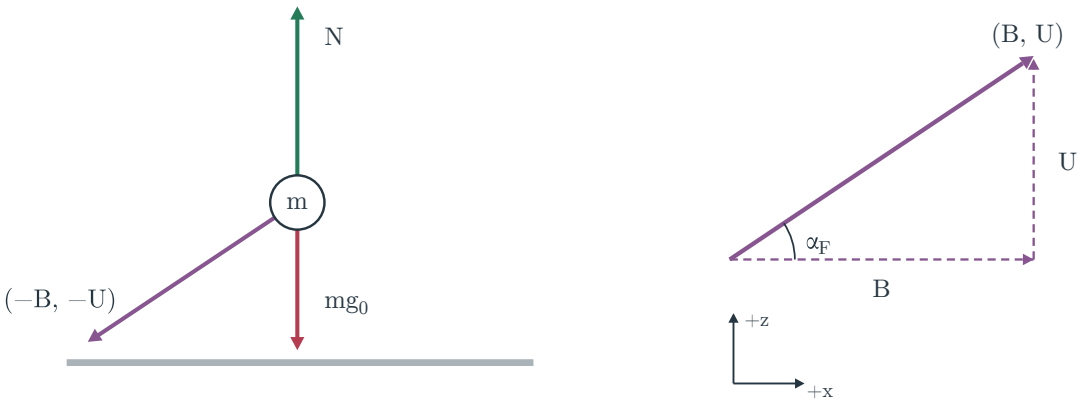


(c) Vertical, positive up along the displayed body axis.

Figure 14: Recovered numerical components from CoasterStats' chest-mounted Mako recording.[31] Opaque segments use confirmed display readings; pale segments show interpolation for visual continuity only. Time identifies source video frames, not original sensor samples. The regular 20 frames/s extraction is supplemented by native-frame peak checks. The axes retain the source convention; transforming them into the assumed seat frame remains a modelling hypothesis. These are motion observations, not measured restraint forces.

direction is held fixed during each structural load path; real pad-normal tracking with arm rotation has not been demonstrated.

Figure 15 separates the two bodies to avoid adding an action–reaction pair to one force balance. Only the forces in panel (a) act on the occupant; panel (b) shows the load passed to the structural model. Positive vertical specific force requires $N - U > 0$, so the seat reaction exceeds the downward pad reaction. It does not require the pad reaction to vanish. The diagram treats the occupant as a point mass and does not locate real contact patches or establish body moment balance, friction, leg compliance or pressure distribution.



(a) Forces on the idealised occupant.

(b) Load on the bar and its components.

Figure 15: Seated-braking hypothesis. In (a), point mass m receives the upward seat force N, weight mg_0 and the pad force $(-B, -U)$. In (b), the equal and opposite force (B, U) acts on the bar; dashed arrows resolve it into components. The angle α_F is measured above the forward horizontal. The grey line represents the ideal frictionless seat. Geometry and arrow lengths are schematic.

For example, $m = 150$ kg, $q_x = -1$, $q_z = +1$ and $\alpha_F = 60°$ give $B = 1.472$ kN, $U = 2.549$ kN and $N = 4.020$ kN. The bar resultant is 2.943 kN, twice body weight, despite positive vertical specific force. This is a hypothetical braking case, not $-2g_0$ airtime. Figure 15(a) explains the larger seat force: it balances both weight and the pad's downward reaction. For $\alpha_F = 30°$ or $45°$ at the same braking magnitude, the bar resultant is instead 1.155 or 1.414 times body weight.

Figure 16(a) plots the longitudinal reading q_x horizontally against the vertical reading q_z vertically. Each point uses both readings from one identified video frame. It is not a time history: point order and separation on the page do not represent elapsed time. Points left of $q_x = 0$ have negative longitudinal specific force; points below $q_z = 0$ have negative vertical specific force. Only points in the lower-left quadrant enter the direct upward-and-forward force assignment. The large cloud above $q_z = 0$ mainly describes positive vertical loading, not an equally large force trying to open the restraint.

Panel 16(b) changes the axes from specific force to force on the bar, in kN, for an assumed 150 kg occupant. Red points assign the opposite components of the lower-left quadrant to the bar. Purple points use the seated-braking hypothesis at $\alpha_F = 60°$ and retain a nonnegative seat reaction. They lie on one straight ray because their ratio $U/B = \tan 60°$ is imposed by that contact

assumption. The horizontal and vertical plotting scales differ, so the physical angle is calculated from U/B, not measured from the printed line. Positive q_z readings can contribute to these purple points while the seat remains loaded. The two point sets are alternative contact constructions, not forces that should be added.

The three curves in panel (b) give model threshold forces for contact positions $s = 0$, 125 and 250 mm, measured from the tube–pad junction along the ideal 250 mm contact segment. For each force direction and contact position, the plotted threshold is the smallest among the seven closure-pair models. To compare a point with a curve, draw a ray from the origin through the point. The curve intersection supplies the threshold's force components in that same direction; calculate the resultant from the axis values. A point between the origin and that intersection has a smaller resultant force than the threshold. These curves mark an assumed nominal yield stress in an included section, not a measured failure boundary.

15.6. Comparison with the reconstructed structural model

The baseline retains all seven conditional closure pairs, $s = 0, 125, 250$ mm, equal branch stiffnesses of 88.7 kN m/rad, and independent pawl/tube yield assumptions of 355 MPa. Here s is the distance from the tube–pad junction E along the ideal 250 mm contact segment, selected independently of the prescribed force direction α_{F}. Selected masses are 100, 125, 150 and 200 kg. Mass is varied independently of closure; no body-size-to-click or dispatch relationship is inferred. Lateral components are retained in the recovered record but excluded from the planar structural calculation.

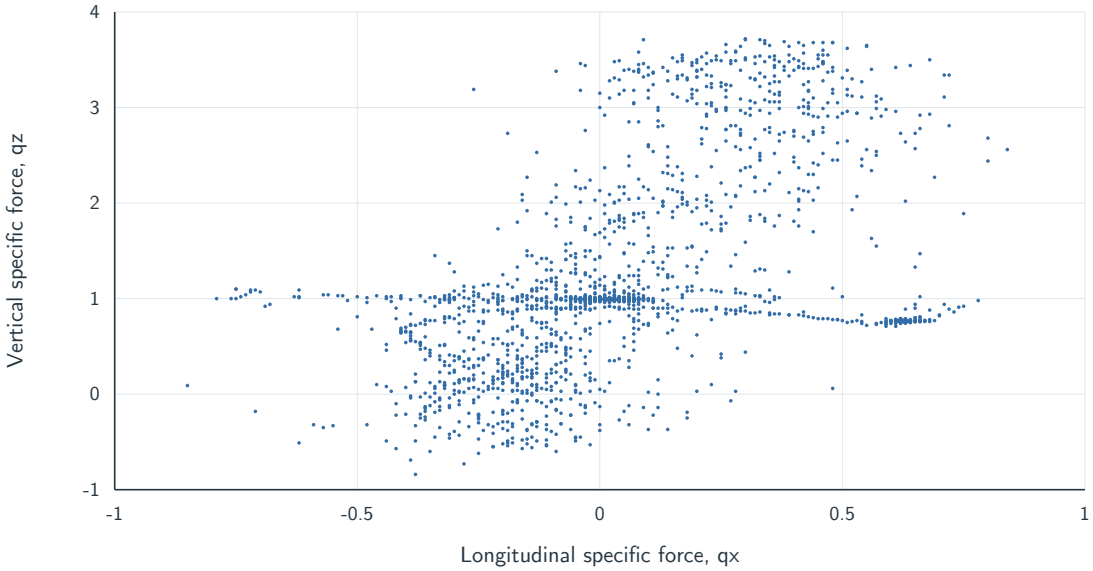
For a nonzero prescribed bar vector, let $F = \sqrt{B^2 + U^2}$ be its resultant force and $\boldsymbol{d} = (B, U)/F$ its direction. For closure pair j and contact position s, define $F_j^*(\boldsymbol{d}, s)$ as the applied force at which a sampled pawl section or the ideal tube first reaches its assumed nominal yield stress along the model's increasing-load path. The reference yield stress is 355 MPa for both submodels. The comparison divides the force inferred under a contact hypothesis by this threshold force for the same direction and configuration:

$$\rho_j = \frac{F}{F_j^*(\boldsymbol{d}, s)}. \tag{33}$$

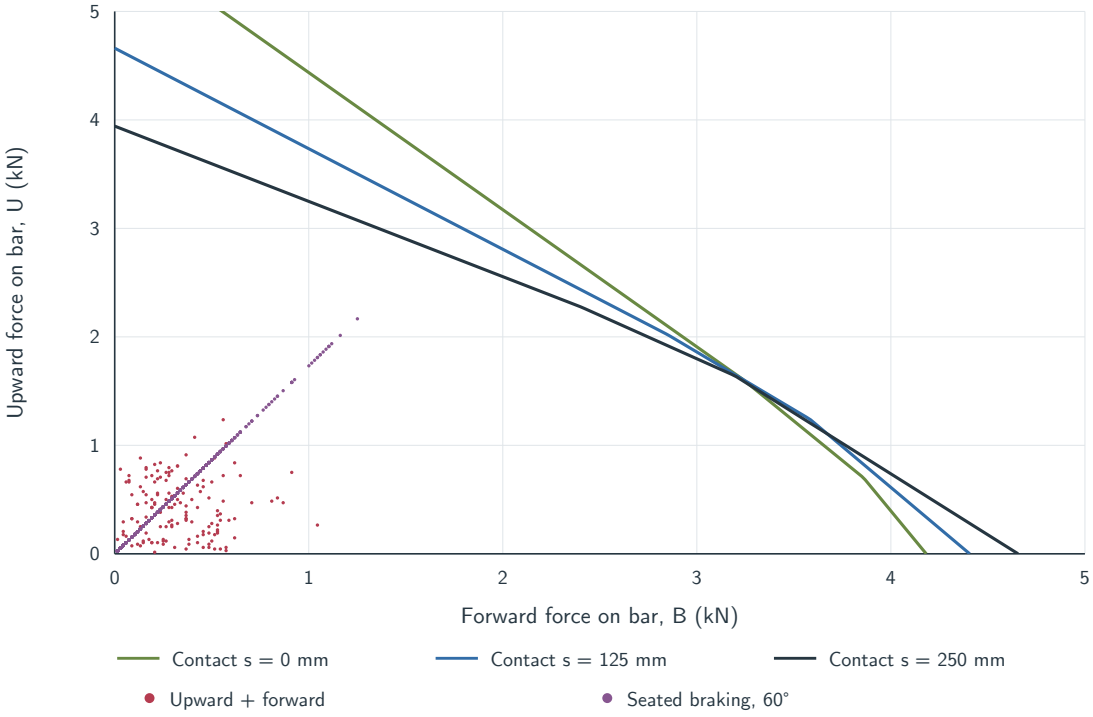
Thus $\rho_j < 1$ means that the applied force is below the model threshold, $\rho_j = 1$ reaches it and $\rho_j > 1$ exceeds it. The force direction is held fixed in each comparison; changing direction, contact position or closure pair can change the denominator. This is a force/threshold-force ratio, not the stress utilisation at the demand force: lever arms and branch allocation evolve along the assumed structural path. Its inverse is not a physical safety factor. Each retained video frame defines a separate quasistatic load-path comparison; the calculation does not integrate a changing-direction, loading/unloading or impact history through the mechanism.

All 135 distinct required directions were evaluated directly with the unchanged structural solver for each of the seven pairs and three contact positions, giving 2835 directional criterion solves. Every feasible confirmed frame was then compared across the four selected masses. The resulting 240744 force comparisons use no angular interpolation or candidate-ranking shortcut. Required missing components are excluded; completing the presentation curves does not enlarge this numerical evidence set.

Table 18 gives the largest applied-force/threshold-force ratio within each contact hypothesis, maximised over the seven pairs and three contact positions. The listed upward-only and upward-



(a) Confirmed same-frame longitudinal and vertical display readings.



(b) Assumed forces on the bar for 150 kg, compared with nominal yield threshold curves for three contact positions.

Figure 16: Motion readings and conditional restraint forces. (a) Each point pairs components from one video frame. (b) Red: direct upward-and-forward assignment; purple: seated braking at 60° for 150 kg. Curves give the lowest nominal yield threshold among seven closure pairs for each contact position. Compare point and curve along a ray from the origin; the plotted axis scales differ. Contact forces and directions are assumptions, not measurements.

Table 18: Largest applied-force/threshold-force ratio in each contact scenario for $m = 150\,\mathrm{kg}$, seven closure pairs and $s = 0, 125, 250\,\mathrm{mm}$. F is the resultant force on the bar; F^* is the force at the first assumed nominal yield stress for that direction and configuration; $\rho = F/F^*$. Braking angles are force directions above the forward horizontal, not restraint closing angles. t_v is source video presentation time, not an instrument sample time.

Case	t_v (s)	B (kN)	U (kN)	F (kN)	F^* (kN)	ρ
Upward only	83.567	0.000	1.236	1.236	3.943	0.313
Upward + forward	83.567	0.559	1.236	1.357	3.294	0.412
Braking, 30°	93.067	1.251	0.722	1.444	3.519	0.410
Braking, 45°	93.067	1.251	1.251	1.769	3.292	0.537
Braking, 60°	93.067	1.251	2.166	2.502	3.251	0.770

plus-forward peaks occur at video time 83.567 s, with the confirmed same-frame components $q_x = -0.38$ and $q_z = -0.84$. All three listed seated-braking peaks instead occur at 93.067 s, with $q_x = -0.85$ and $q_z = +0.09$. These recovered extrema are 9.500 s apart. Combining the separately most negative longitudinal and vertical values would construct an unobserved vector. The direct upward-and-forward assignment nevertheless tests every retained frame in which both components are negative. Other closure/position choices can select other critical frames; the table does not identify one universal critical instant for every model.

For the selected 150 kg mass, the most demanding comparison is seated braking at $\alpha_\mathrm{F} = 60°$, $s = 250$ mm for the final conditional pair A38/40+B40. The prescribed resultant force on the bar is 2.502 kN, compared with a nominal yield threshold force of 3.251 kN for that direction and configuration. Their quotient, $2.502/3.251 \simeq 0.770$, means that the applied force is approximately 77.0% of that threshold force. Under this contact hypothesis, the simultaneous components are $B = 1.251$ kN and $U = 2.166$ kN, while the seat reaction is $N = 2.299$ kN. Thus the seat remains loaded even though substantial opening demand is assigned to the bar; the small positive q_z describes their net vertical action.

At 200 kg, the same frame and reference configuration give an applied resultant force of 3.335 kN against the same 3.251 kN threshold: $3.335/3.251 \simeq 1.026$, or approximately 102.6% of the model threshold force. The assumed force therefore exceeds the first included nominal yield threshold in this calculation. It does not establish physical yielding, fracture or release of the real restraint. Conversely, the selected 100, 125 and 150 kg forces remaining below their corresponding thresholds do not establish service adequacy for all bodies at those masses. Mass, fit and closure have not been linked by an occupant model.

The extraction method materially affects the applied-force/threshold-force ratio F/F^*. At 150 kg, the overview-stroke median method gave 0.217 for upward force alone, 0.302 for the upward-and-forward assignment and 0.668 for seated braking at 60°; confirmed numerical readings raise these force ratios to 0.313, 0.412 and 0.770, respectively. The corresponding increases are approximately 44%, 36% and 15%. The new values are close to, but can exceed, earlier same-column stroke-endpoint constructions. Those endpoints are rendering scenarios, not simultaneous measured vectors or uncertainty bounds. They do not bound the numerical readings.

The written overview minimum $q_x = -1.50$ is not inserted into the confirmed sequence. Combining it with $q_z = -0.85$ would remain a source-label sensitivity construction, not an observed

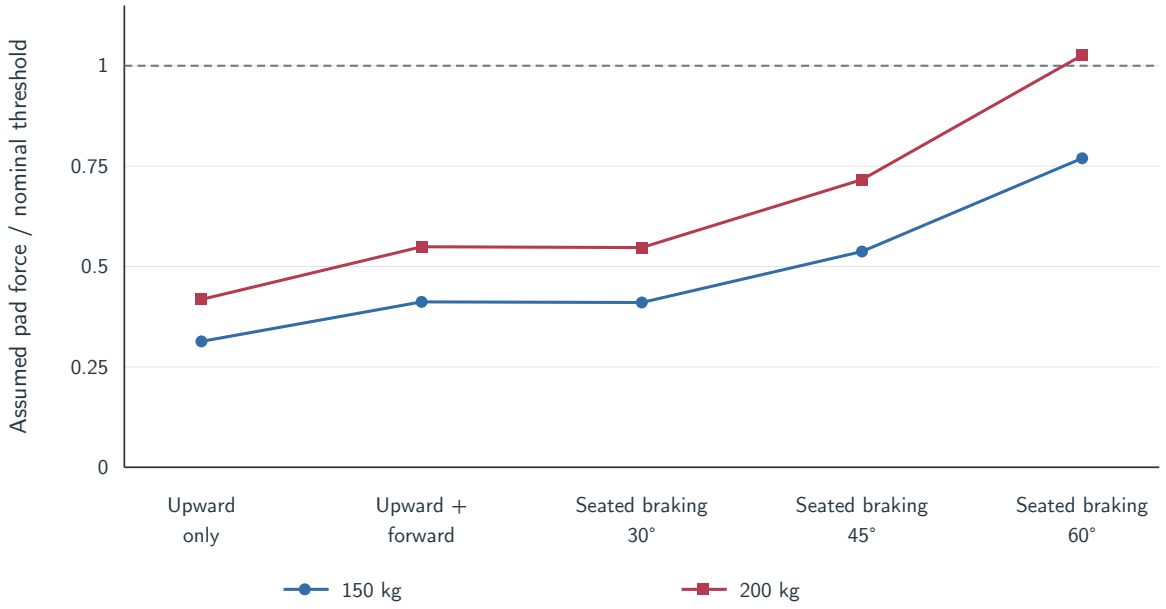


Figure 17: Largest ratio of assumed resultant pad force F to nominal yield threshold force F^* within each contact hypothesis, maximised over the seven closure pairs and three contact positions. The first two cases assign only the upward component, or both upward and forward components, to the bar. The three seated-braking cases prescribe force directions $\alpha_F = 30°$, $45°$ and $60°$ above the forward horizontal; these are not restraint closing angles. The series use assumed occupant masses of 150 and 200 kg. The line at $F/F^* = 1$ marks equality with the model threshold force, not a physical failure boundary or permitted passenger mass.

event. The hypothetical $q_x = -1$, $q_z = -2$, 150 kg example in Section 13.8 also remains separate: its 3.290 kN resultant force is compared with a 3.561 kN model threshold, giving $F/F^* \simeq 0.924$. Neither construction establishes that those components occurred together on Mako.

15.7. Sensitivity, transients and what remains unresolved

The 420 peak summaries cross five contact hypotheses, four masses, three contact positions and seven pairs. A further 455 directed sensitivity cases were repeated using the newly confirmed baseline critical frame for each case and pair at 150 kg and $s = 125$ mm. They vary independent pawl strengths of 355/650/900 MPa, tube strengths of 235/355/460 MPa, the stiffness-scale parameter $\eta = 0.25, 1, 4$ and the branch-stiffness ratio $\kappa = 0.25, 1, 4$, defined in Section 13. The critical frame is not reoptimised for each resistance hypothesis, so these are responses to selected vectors rather than global maxima of every altered model. All 455 selected paths reach an included criterion within the model domain. The choices have no probability weights and do not identify installed materials or compliance.

Contact direction is itself influential. In Equation (32),

$$\frac{\partial U}{\partial \alpha_{\mathrm{F}}} = B \sec^2 \alpha_{\mathrm{F}}, \qquad \alpha_{\mathrm{F}} \text{ in radians.} \tag{34}$$

At 150 kg, $q_x = -0.85$ and 60°, this is about 87 N per degree locally. Improving the pad normal, centre of pressure and seat/body contact geometry is therefore more informative than adding precision to a mass threshold. Unresolved readings, unexamined native frames, display processing and the unknown contact geometry still prevent a complete experimental demand envelope. A nominally consistent force scale does not establish adequacy for every body that can fit.

A quasi-static force comparison also omits free motion across a gap and subsequent contact impact. In a simple undamped spring-impact idealisation, kinetic energy gives $F_{\mathrm{peak}} \simeq v_{\mathrm{rel}}\sqrt{m_{\mathrm{eff}}k}$, with no continued external forcing. This is not an estimate for the restraint: its effective mass, stiffness, gap, initial relative velocity, damping and contact evolution are unknown. It illustrates why one cannot assign a universal amplification factor from an acceleration maximum. The restraint's own weight and inertia, changes of contact, three-dimensional forces, and transmission/anchorage resistance remain necessary parts of an actual dynamic assessment.

The extension supports a limited physical conclusion: the reconstructed model can be compared with sub-kN to few-kN force scales derived from published ride motion under explicit contact hypotheses, and braking can load the bar while the seat remains loaded. It does not establish the real assembly's maximum force, a permitted passenger mass, or a family-wide acceleration limit. Useful next evidence would be the original synchronised acquisition with its processing and mount geometry, measured pad/seat surfaces and contact locations, and verified transmission, support and material properties. Other Hypers provide comparison data; their appearance and product-family membership do not authenticate the same mechanism revision.

16. Discussion

16.1. What the equilibrium result establishes

The main result is a traceable connection between reconstructed contours, closing positions, unilateral contact and load transmission. All nine working cases admit an in-plane rigid-body

equilibrium with compressive normal reactions and no imposed residual pawl counter-torque. This depends on finding the seated configuration: a force at first touch with the pawl held fixed may generate a moment that disappears when the pawl is allowed to rotate into its load-bearing configuration. A normal drawn on an arbitrary pose is therefore insufficient to establish retention.

The analysis also distinguishes an equivalent force from a real contact area. Contact reactions represent resultants of distributed tractions. At an ideal corner, two generating normals satisfy equilibrium and describe an admissible reaction cone, but they do not determine curvature, deformation or peak pressure. At B–B 32°, keeping both separated contact points is essential when calculating section moments. Replacing them by a force at a visually selected point without its transport couple would lose information.

Non-penetration, reaction signs, equilibrium and coordinate transformations support the internal consistency of the formulated problem. They do not experimentally validate the contours. Tolerance, radii, wear or tracing error could change the configuration. Numerical precision must remain distinct from measurement accuracy.

16.2. Physical redundancy, load paths and shared components

Four identified pawls, several load-bearing contact reactions and independence against failure are different concepts. One pawl can sustain two normal reactions; the two pawls on one side can engage at successive positions; and the two sides can receive different moments. None of these observations justifies multiplying one component's resistance by four or assigning each pawl one quarter of the load.

The main shaft 5832 is shared. The arm, connections, carrier, supports and seat frame also belong to the transmission chain. Distinct contact branches can therefore retain common structural dependencies. Release adds another functional network whose complete axial arrangement has not been reconstructed. Different sector and foreground-pawl references document different designs; they do not establish statistically independent failures, wholly separate actuation or tolerance of every common-component failure.

Assigning the full moment to one side is a useful static scenario. Interpreting it as residual strength after a failure would also require demonstrating that the remaining branch can reach a compatible contact, that its supports and connections remain intact, and that recovery motion does not introduce a different dynamic demand. A local loss can alter alignment, clearance or position elsewhere. A post-failure state cannot be obtained simply by doubling a force in an intact state. Failure probabilities and physical-system redundancy are not established here.

16.3. Compatibility and moment allocation

Shaft continuity couples the two views. The interleaved nominal angles are consistent with staggered engagement, but they do not resolve which branch first carries opening load at each global arm position. Initial clearances and flexibility of shaft, sectors, pivots and frame determine whether multiple branches become active together. One reaction may develop before another gap closes.

The envelope retains this uncertainty through α: $M_A = \alpha M$ and $M_B = (1 - \alpha)M$ prescribe an allocation rather than a measured distribution. Its 5625 evaluated responses are independent evaluations, not 5625 complete assembly simulations or validated pairs of poses. Section 13 adds a reduced common-arm network in which moment allocation follows assumed transmission compliance

and reconstructed onsets. That rotational construction does not establish full three-dimensional contact and deformation compatibility in the physical assembly.

16.4. Conditional resistance and physical capacity

The reconstruction calculates internal demand for an assigned moment and permits geometry or material scenarios to be compared. Nineteen of 27 sampled pawl sections were accepted; eight were excluded geometrically. For B–B 40°, the largest nominal value among accepted sections is approximately 129 MPa at 16 mm thickness and 1000 N m. It is neither the part's local maximum, an allowable stress nor a capacity test. Different sampled regions also prevent directly ranking the real pawls by the largest accepted value.

Pins, fasteners and tube models have similarly specific scopes. The chosen fastener model conserves force and eccentric moment but does not identify preload, fit or actual face transfer. Pin diameters and spans are parameters. The straight tube beam locates a nominal maximum within that idealisation, excluding joints, welds, discontinuities and actual support compliance. Even within the ideal beam, a small shaft moment does not imply small bending everywhere: the pad-side end can carry a larger section moment.

Global capacity requires comparing relevant resistance modes with a defined demand and checking the connections between them. Effective radii, supports, materials and treatments, fits, surface condition, body-load distribution and load histories remain incomplete. A static nominal stress does not determine fatigue life; its ratio to an assumed yield strength is not a global safety factor. Current responses help identify which missing data matter most.

16.5. External geometry and uncertainty

The photographic estimates of 780 mm, 98° and 250 mm define an equivalent arm and contact segment. Scaling from an assigned 32 mm shaft bore and recovering approximately 60 mm at the tube provides local consistency, not independent physical calibration or guaranteed scale over the full arm. The pad is curved and three-dimensional; planar rectification does not automatically recover its load-bearing surface or offset from the tube.

For a normal force following the ideal segment, the effective lever is $d = -L\cos\beta + s$. Locally,

$$\Delta d \simeq -\cos\beta\,\Delta L + L\sin\beta\,\Delta\beta + \Delta s, \qquad \Delta\beta \text{ in radians.}$$

At $L = 780$ mm and $\beta = 98°$, angular sensitivity is approximately 13.48 mm per degree; length sensitivity is 0.139 and centre-of-pressure sensitivity is one. This explains why improving the angle and line of action may be more useful than adding decimal places to length. It does not provide an error interval without measured uncertainties and correlations.

The 125 external cases and five allocations form a selected deterministic grid. Its largest result is not a measured acceleration or a test load. Tangential ratios specify force directions rather than measured friction. Closing moments retain their sign instead of being converted into opening by an absolute value. A physical service demand would require passenger interaction with seat, backrest and pad, vehicle motion, and the restraint's own inertia.

16.6. What the ride measurements change

The extension in Section 15 replaces purely chosen acceleration values with attributed observations for some demand comparisons. It preserves their limitations: the Mako record is chest-mounted,

some numerical readings remain illegible and its summary extrema are not wholly consistent with the onride display. Identified source-frame timestamps improve video timing without recovering acquisition timestamps or sensor/video synchronisation. The other reviewed Hyper measurements add experimental context without authenticating an identical mechanism revision or complete service envelope.

The contact hypothesis changes the mechanical interpretation. Negative vertical specific force can motivate an upward bar-force comparison, while positive vertical specific force can coexist with upward bar load during braking because the seat and pad reactions oppose each other. A measurement of their net action does not identify either contact separately. The most demanding selected comparison assumes seated braking with force 60° above the forward horizontal at the outer contact position. Its calculated bar force is 2.502 kN for an assumed occupant mass of 150 kg and 3.335 kN for 200 kg, compared with a 3.251 kN nominal yield threshold force for the same model direction and configuration. Dividing each force by that threshold gives 0.770 and 1.026, respectively. Thus the former is below the model threshold and the latter above it; neither establishes a real operating margin or physical failure. The contact geometry, input uncertainty and omitted resistance modes cannot be resolved by choosing the more reassuring curve.

The quasistatic structural paths are independent comparisons for the prescribed vectors, not a simulation of the complete ride history. They do not model body motion, clearance impact, unloading, branch switching or retention after local deformation. Original synchronised data and an identified occupant-contact model would allow a more informative next comparison than increasing nominal numerical precision.

16.7. Documentation and transfer to other installations

Patents describe embodiments, photographs show particular units, and testimony records an individual's experience. None automatically replaces manufacturing drawings or establishes vehicle configuration and revision. A part reference does not authenticate material, date or deployment across other trains.

This study does not determine the cause of the Goliath accident. A similar architecture or a nominal stress compatible with a steel scenario cannot reconstruct a seat's condition during an event. Fracture, release, supervision and passenger containment require different evidence. Attribution of photographs to Mako and extrapolation to other Hyper installations retain their documented limits.

17. Conclusions and further work

A documentary and geometric model has been developed linking component identification, closing sequences, local seated contacts and documented load scenarios. Each view contributes four nominal click positions, interleaved from 24° to 40° in the nominal open-reference frame. A–A 40° adds no fifth distinct tooth configuration; A9 remains a possible indexing device. Release is described separately from incremental retention equilibrium.

Nine studied cases admit frictionless in-plane equilibrium and locally block opening. Reaction coefficients and nominal stresses propagate chosen external loads with explicit signs, units and exclusions. The functional axial model connects one shaft 5832, two sectors and four pawls; the number and positions of its supports, and the actual moment allocation, remain unresolved.

Conditional nominal load thresholds have also been obtained for upward, forward and combined reference forces, with the CAD orientation, tube-axis offset and mounting hypothesis recorded explicitly. These crossings do not establish fracture or loss of retention. The next priorities are support and axial-transfer dimensions, validating the assumed transmission compatibility, and checking the contact geometry. Radii, fits, material properties and a defined external demand would permit specific resistance checks beyond nominal stress. A reduced common-arm network further illustrates sequential engagement and the dependence of the first included criterion on assumed transmission flexibility. Seven closure-position pairs extend the comparison beyond the final nominal position. Illustrative occupant-load checks show why a selected reference mass is not a passenger limit, while leaving real service demand and margin unverified. The soft-link scenarios require several degrees of transmission twist, so their higher force thresholds depend on an unverified compliance assumption. Ultimate capacity, fatigue life and physical validation remain unresolved.

This revision adds an experimentally informed, reproducible comparison rather than a capacity validation. Confirmed Mako numerical components retain a common identified video frame. Pale interpolation completes the presentation figure, while every force comparison excludes required illegible components. The recovered longitudinal and vertical minima occur 9.5 s apart, so their separate extrema are not combined into a measured demand vector. Explicit contact hypotheses show that the load transmitted to the bar cannot be inferred from a single acceleration peak. Some selected model comparisons approach or cross an included criterion when the contact, mass and resistance assumptions change; the available evidence therefore does not establish a universal passenger mass or operating margin.

Author note and contributions

I write as `u/lucky_3838` and work as a university lecturer and researcher in applied physics, with an engineering background, a PhD and experience in numerical modelling. Visits to European parks and an interest in B&M and Intamin form part of my broader curiosity about forces, mechanisms, launch systems, control and automation. This study grew out of a personal interest in understanding the restraint, outside my professional specialism. No institution is identified or represented here.

I conceived and directed the study, assessed the visual sources, performed the reverse engineering from images and the complete CAD reconstruction, developed the mechanical interpretation, and supplied the dimensional estimates. I directed the calculations and critically reviewed the analysis and manuscript throughout their development.

AI assistance was used for calculations and to improve the clarity of the text.

References

[1] Bolliger & Mabillard. *About us*. https://www.bolliger-mabillard.com/about.

[2] Bolliger & Mabillard. *Hyper Coaster*. https://www.bolliger-mabillard.com/hyper-coaster.

[3] Busch Gardens Williamsburg. *Apollo's Chariot: Exclusive Members Only Ride Event*. 2019. https://buschgardens.com/williamsburg/blog/apollos-chariot-exclusive-members-only-ride-event/.

[4] Six Flags Great America. *Raging Bull*. https://www.sixflags.com/greatamerica/attractions/raging-bull.

[5] Bolliger, Walter; Mabillard, Claude. *Installation for amusement park*. 2001. US patent US6287211B1. https://patents.google.com/patent/US6287211B1/en.

[6] Jaskol, James K.; Tomlin, Jason C.; Athans, Kristin Lynn; Wong, Kwok Wah James; Yeh, Edward. *Pawl and ratchet assembly adapted for indirect sensing of proper operation*. 2011. US patent US7947943B2. https://patents.google.com/patent/US7947943B2/en.

[7] Lubman, Phillip J.; Ramirez, Sean A.. *Lap bar assembly with locking mechanism with locking in lap bar and grab bar positions*. 2012. US patent US8132514B2. https://patents.google.com/patent/US8132514B2/en.

[8] Greathead, E.; Taylor, J.; Rai, J.. *Rollercoaster Restraint vs Your Spine*. 2024. Journal of Physics Special Topics. https://journals.le.ac.uk/index.php/pst/article/download/4796/4821/17345.

[9] Valente SAS. *Fabrication*. https://www.valentesas.fr/fabrication.html.

[10] Valente SAS. *Présentation*. Company brochure, one page. https://valentesas.fr/wa_files/Presentation_20Valente.pdf.

[11] Office of the Commissioner of Insurance and Safety Fire, State of Georgia. *Commissioner King Provides Update on Extensive Investigation into Six Flags Tragedy*. 2026. https://oci.georgia.gov/press-releases/2026-10-15/commissioner-king-provides-update-extensive-investigation-six-flags.

[12] SeaWorld Orlando. *Accessibility Guide*. Park guide, May 2026, 32 pages; fit requirements on pages 11–14 and Mako entry on page 22. https://seaworld.com/orlando/-/media/commercial/seaworld-orlando/files/pdfs/swo_accessibility_guide_may_2026-1.pdf.

[13] Russ Tedrake. *Robotic Manipulation, Ch. 5: Bin Picking*. https://manipulation.mit.edu/clutter.html.

[14] Richard T. Barrett. *Fastener Design Manual. NASA RP-1228, 1990*. https://ntrs.nasa.gov/citations/19900009424.

[15] MIT OpenCourseWare. *1.050 Solid Mechanics: Shafts in Torsion, 2004*. https://ocw.mit.edu/courses/1-050-solid-mechanics-fall-2004/8c30fdf15d9d50c02f4215903c847d7f_emech8_04.pdf.

[16] COMSOL. *Cylinder Roller Contact, version 6.3 documentation*. https://doc.comsol.com/6.3/doc/com.comsol.help.models.sme.cylinder_roller_contact/cylinder_roller_contact.html.

[17] Jared A. Grauer; Matthew J. Boucher. *Output Measurement Equations for Flexible Aircraft Flight Dynamics. NASA/TM-2018-220102*. https://ntrs.nasa.gov/citations/20180007917.

[18] OpenStax. *University Physics, vol. 1, Section 10.6: Torque*. https://openstax.org/books/university-physics-volume-1/pages/10-6-torque.

[19] OpenCV. *Basic concepts of the homography explained with code*. https://docs.opencv.org/5.0/tutorials/features/homography/homography.html.

[20] Ryan The Ride Mechanic. *Goliath ride accident, speculation, and thoughts*. 2026. YouTube video, identified by its on-screen title and channel in video captures.

https://www.youtube.com/watch?v=kb7oDUWhSD4. At approximately 5:44, the speaker credits pictures supplied by @Crazybird199, also credited on screen. The original video page could not be independently revalidated. Used to identify the provenance of visual material, not as a finding about the incident.

[21] Velocity Rides. *Photograph post in Roller Coaster Enthusiasts Club*. Facebook, undated. https://www.facebook.com/photo?fbid=122191604276972360&set=pcb.28799588739690359. Source identified through a capture of the post; the live page could not be retrieved. The original photographer has not been identified. Cited for photographic provenance, not for the accompanying incident claim.

[22] ThrillZing. *Mako's lift hill, first drop and camelback hills across the lake in the late afternoon, SeaWorld Orlando*. Photograph, 10 March 2021. Wikimedia Commons (original file page). Photographer's own work, licensed under Creative Commons Attribution 4.0 International.

[23] Böllhoff Group. *Product data sheet: ISO 4017 similar to DIN 933, steel 8.8 hexagon head bolts*. Dimensional table, p. 1; M16 head width across flats: 24 mm. https://eshop-tr.boellhoff.com/out/media/pdf/ISO_4017_Stahl_8.8_zdpb1___en.pdf. Catalogue comparison only; not an identification of the installed bolts.

[24] Europa-Park. *Technische Daten Silver Star*. 20 March 2002. https://presse.europapark.com/de/presse/nachricht/datum/2002/03/20/technische-daten-silver-star/?type=2. Historical park technical sheet; the airtime figure is not a measured load history or a family-wide design envelope.

[25] Manuel Schüttler and Thomas Wilhelm. *Bewegungsanalyse im Freizeitpark*. Praxis der Naturwissenschaften – Physik in der Schule 60(6), 18–24, 2011. https://www.thomas-wilhelm.net/veroeffentlichung/freizeitpark.pdf. Silver Star measurements and interpretation of accelerometer readings, Section 4.2.

[26] SSAB. *SSAB Tubes and Sections*. Product brochure, 2019, printed p. 5: structural tube grades and mechanical properties. https://www.ssab.com/-/media/files/en/tubes-and-sections/802_specialists-in-tubes-and-sections-en.pdf. Generic material context, not identification of the restraint tube.

[27] Ovako. *42CrMo4: material data sheet*. Steel Navigator. https://steelnavigator.ovako.com/steel-grades/42crmo4/. Product variants and condition-dependent properties; no attribution to B&M parts.

[28] Böllhoff. *Manual of Fastening Technology*, 8th edition. Table 3.2, printed p. 24: mechanical properties of steel screws. https://eshop-tr.boellhoff.com/out/media/pdf/the_manual_of_fastening__8100-en.pdf.

[29] International Organization for Standardization. *ISO 898-1:2013: Mechanical properties of fasteners made of carbon steel and alloy steel—Part 1*. Published scope and abstract. https://www.iso.org/standard/60610.html. Cited for the scope of the property-class standard; no joint-capacity verification is claimed.

[30] Rosemarie Figueroa Jacinto, Rachel Hybart and Jessica Zendler. *Applications of Motion Capture in Amusement Ride and Devices Assessments: Exploring Its Role in Evaluating Rider Interaction, Comfort, and Safety*. Proceedings of the Human Factors and Ergonomics Society Annual Meeting 69(1), 568–572, 2025. https://doi.org/10.1177/10711813251367744.

[31] CoasterStats. *Mako g-force onride – SeaWorld Orlando [4K 60FPS]*. Video, 7 September 2025. https://www.youtube.com/watch?v=7z4nijr7ExM. Chest-mounted recording; overview chapter at 2:32. Numerical display readings and identified source-frame presentation times are reconstructed in this revision; original acquisition data were not recovered.

[32] Manuel Schüttler. *Physik im Freizeitpark: Möglichkeiten und Vergleich von Beschleunigungs- und GPS-Messung sowie Videoanalyse*. Written examination thesis, Julius-Maximilians-Universität Würzburg, February 2011; supervisor Thomas Wilhelm.

https://www.thomas-wilhelm.net/arbeiten/Physik_Freizeitpark.pdf. Instrument descriptions on printed pp. 6–10; Silver Star on pp. 95–98; companion-DVD inventory in the appendix.

[33] CoasterStats project discussion, with first-person team replies by SignatureSpare. *Rollercoaster G-force Measurements [Other].* 2020. https://www.reddit.com/r/rollercoasters/comments/jddt2l/rollercoaster_gforce_measurements_other/. Historical processing and research-sharing account; not a Mako-specific acquisition protocol.

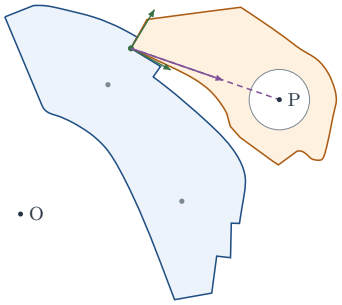
[34] Trax (CoasterStats). First-person methods account, post 273 in *Freizeitpark Influencer*, 6 February 2026. https://epfriends.de/index.php?postID=459722&thread%2F4001-freizeitpark-influencer%2F=. General acquisition, mounting and processing account, not an independent calibration certificate.

[35] 0Kajuna0 / App Prototyping Barcelona. *Roller coaster forces in Ferrari Land / Port Aventura.* First-person instrument discussion, 30–31 May 2017, posts 3, 5 and 11. https://coasterforce.com/forums/threads/roller-coaster-forces-in-ferrari-land-port-aventura.41137/. Accompanying video: https://www.youtube.com/watch?v=bg8B0Hl_au0. Mouth-mounted camera measurements and processing description; no validated pad-load data.

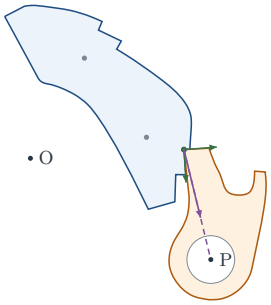
[36] Busch Gardens Williamsburg. *Launch into Physics.* Official educational workbook, undated. https://buschgardens.com/williamsburg/-/media/commercial/busch-gardens-williamsburg/files/youth/physics-day/lip-workbook.pdf. Accuracy disclaimer on p. 14; Apollo's Chariot section on pp. 111–118. Not adopted as measured demand validation.

[37] Six Flags Great Adventure. *Great Adventure Physics.* Official educational workbook, 2024, p. 13. https://static.sixflags.com/website/files/SFGAD%20Physics%20Workbook%202024.pdf. Suggested classroom factors do not constitute an acquired Nitro time series.

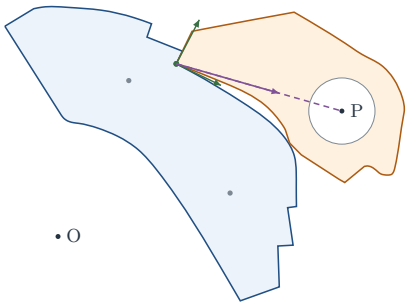
Appendix A. Seated contact configurations



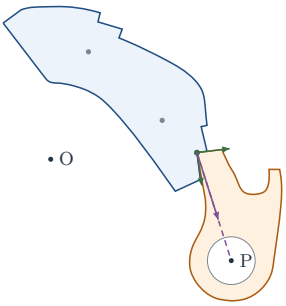
(a) A–A 24° (A2); $R/M = 13.061$.
$\delta = 1.197°$, $\phi = 0.571°$.
$(N_1, N_2)/M = (12.781, 2.132)$.



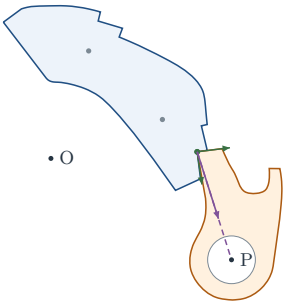
(b) A–A 29° (A3); $R/M = 12.129$.
$\delta = 0.943°$, $\phi = 0.669°$.
$(N_1, N_2)/M = (11.970, 2.134)$.



(c) A–A 33° (A2); $R/M = 13.902$.
$\delta = 0.928°$, $\phi = 0.467°$.
$(N_1, N_2)/M = (13.620, 2.380)$.



(d) A–A 38° (A3); $R/M = 12.987$.
$\delta = 0.982°$, $\phi = 0.812°$.
$(N_1, N_2)/M = (12.769, 2.115)$.

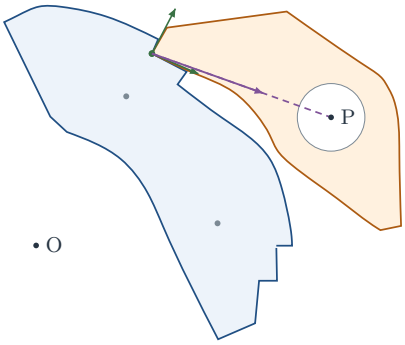


(e) A–A 40° (A3); $R/M = 12.987$.
$\delta = 2.509°$, $\phi = 0.589°$.
$(N_1, N_2)/M = (12.769, 2.115)$.

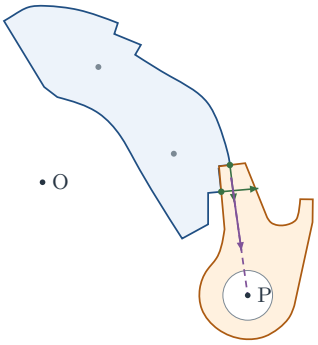
Toothed sector — Active pawl
Normal forces on pawl — Resultant line

O: main rotation centre. P: pawl pivot.
The purple line passes through P.
The 38° and 40° cases reach the same tooth configuration.

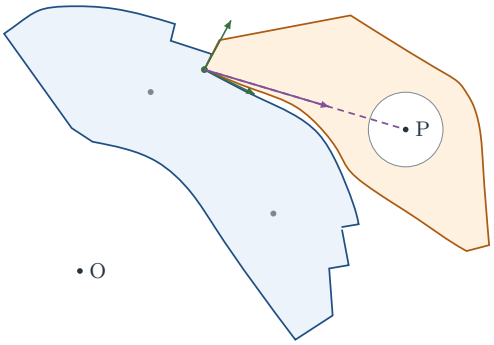
Figure A.1: Reconstructed A–A contact configurations. δ and ϕ are adjustments from the source poses, not measured backlash. Reaction coefficients R/M and N_i/M are in N/(N m). The arrows show contact forces on the pawl, opposite to the sector normals in Section 8; arrow lengths are schematic. Each panel is scaled independently; grey dots mark fixing centres.



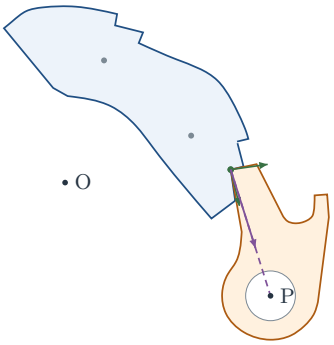
(a) B–B 27° (B2); $R/M = 12.823$.
$\delta = 1.048°$, $\phi = 0.475°$.
$(N_1, N_2)/M = (0.882, 12.730)$.



(b) B–B 32° (B3); $R/M = 11.040$.
$\delta = 0.549°$, $\phi = 0.618°$.
$(N_1, N_2)/M = (11.032, 0.206)$.



(c) B–B 36° (B2); $R/M = 13.607$.
$\delta = 0.754°$, $\phi = 0.565°$.
$(N_1, N_2)/M = (2.214, 13.329)$.



(d) B–B 40° (B3); $R/M = 12.803$.
$\delta = 0.302°$, $\phi = 0.328°$.
$(N_1, N_2)/M = (0.892, 12.688)$.

Toothed sector Active pawl
Normal forces on pawl Resultant line

Figure A.2: Reconstructed B–B contact configurations. The 32° case retains two separate contact points; the other cases combine normal directions at an ideal corner. Reaction coefficients are in N/(N m), with M assigned to each case independently. Arrows show contact forces on the pawl, opposite to the sector normals in Section 8; their lengths are schematic, not a pressure footprint.

Appendix B. Selected source video frames

The five frames reproduced here document visual material used to interpret the locking assembly and reconstruct its geometry. They are screen captures from Ryan The Ride Mechanic's *Goliath ride accident, speculation, and thoughts.*[20] The presenter inset, player overlays and visible credits are retained as captured. Four frames explicitly display the credit `Pictures from @Crazybird199`; the video also identifies that subscriber as the supplier of pictures. This records the contribution and route into the video, without establishing the original photographer or authorship of the technical drawing.

The drawing capture and photographs provide complementary views, with different limitations in perspective, resolution and occlusion. They do not independently authenticate the ride or assembly revision, identify materials, or establish load capacity. Their inclusion does not adopt the video's incident commentary. Readable analytical geometry is presented in the reconstructed figures in the main text.

Figure B.1 records the sheet and the relationship between its section views. Figure B.2 connects the reconstructed profiles to loose components and an oblique assembly view. Figure B.3 provides the wider installation context and a closer view of the central linkage. Together they expose which features are visible in the source material and which connections require interpretation.

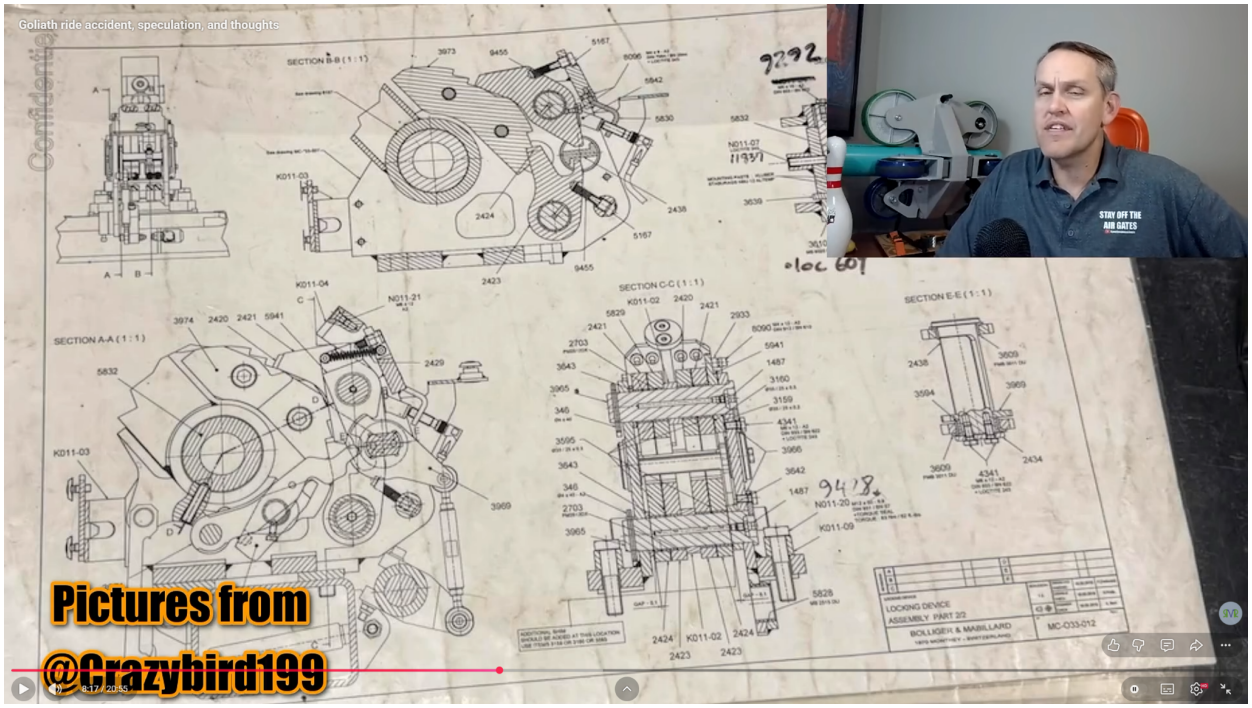
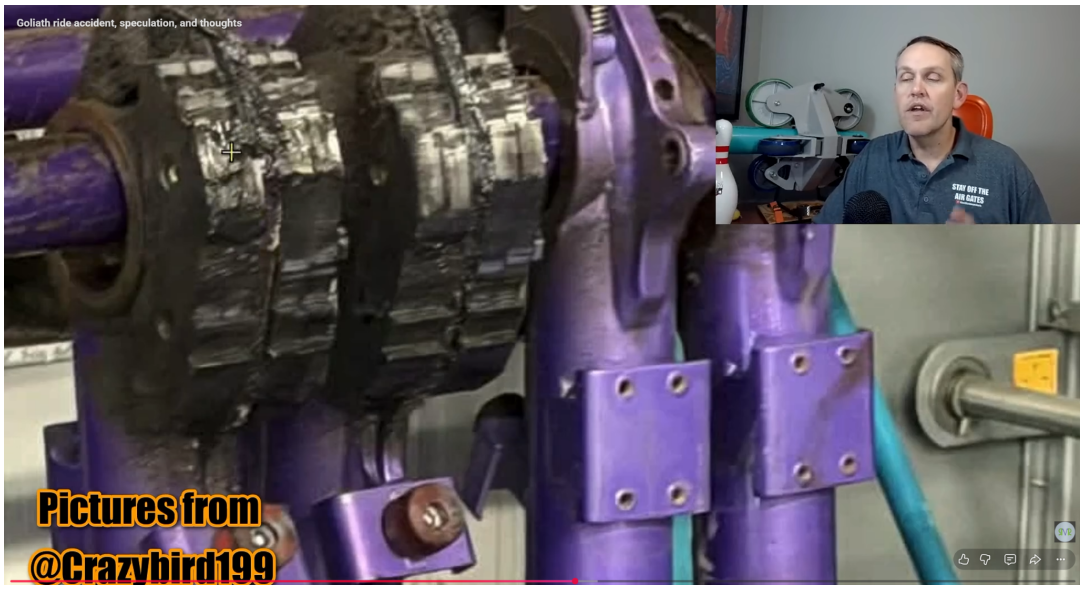


Figure B.1: Assembly drawing displayed in the video at 8:17. Sections A–A and B–B and complementary assembly views are visible, together with a Bolliger & Mabillard title block and the marking *Confidentiel*. The on-screen picture credit names `@Crazybird199`.[20] The printed *1:1* applies to the original sheet; this screenshot has no authenticated reproduction scale.

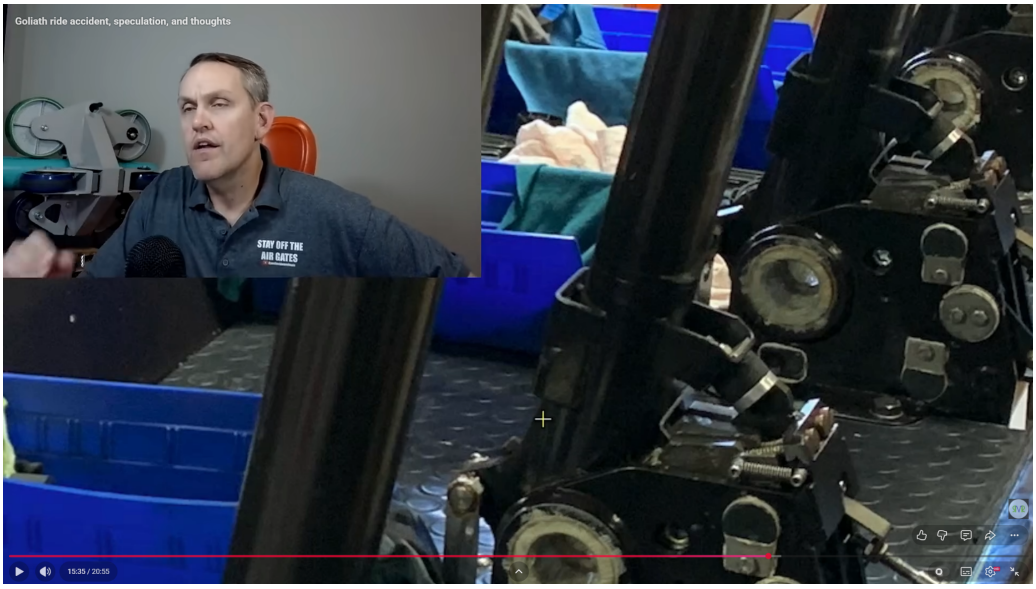


(a) Loose toothed sectors and a pawl on a workbench.

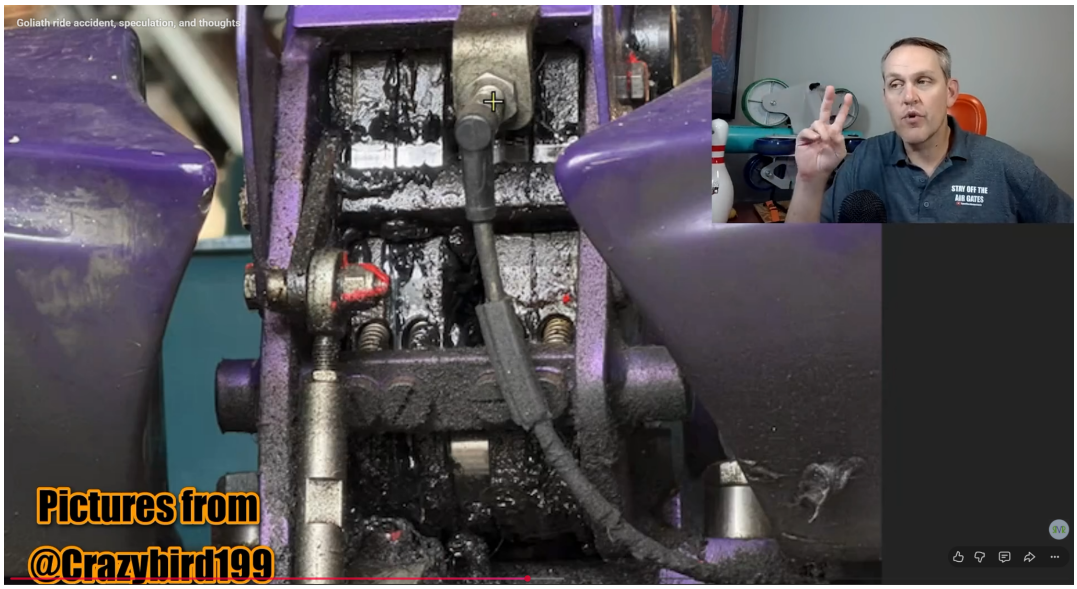


(b) Oblique view of exposed toothed members, supporting plates and adjacent shaft components.

Figure B.2: Component and assembly views shown in Ryan The Ride Mechanic's video; both frames retain the explicit picture credit to `@Crazybird199`.[20] Profiles, bores and the axial arrangement can be compared qualitatively with the reconstruction. These views alone do not establish dimensions, fits or the load carried by each visible member.



(a) Exposed mechanisms along a row, with tubular arms and spring-loaded linkages visible (15:35).



(b) Close view between surrounding structures, showing bolted pivots, springs and a central linkage.

Figure B.3: Further mechanism views from Ryan The Ride Mechanic's video.[20] Panel (b) retains the explicit on-screen picture credit to `@Crazybird199`; that credit is not visible in panel (a). The presenter and captured overlays are retained. The photographs provide spatial and component context, rather than a complete dimensioned assembly or a verified support model.